

Inferring Human Whole-Brain Effective Connectivity from Resting-State fMRI Using Linear Least Squares Regression

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Abstract

The release of large-scale fMRI datasets, such as the Human Connectome Project resting-state fMRI has enabled higher resolution, longer duration observations of brain activity than previously possible. This creates a new opportunity to apply scalable inference methods, such as linear least squares (LLS) regression, to estimate directed effective connectivity (EC) across the entire brain. Rigorous evaluation was conducted on synthetic networks emulating fMRI-like dynamics. These tests explored the effects of sampling size, and intrinsic connectivity, revealing conditions under which group-level EC can be reliably recovered. Applied to the HCP dataset, the method produced a whole-brain EC across 333 cortical regions. The resulting network reveals large-scale functional components, including dominant excitatory pathways and opposing high and low order functional systems. Further refinement using a delay based model uncovered the lagged coupling effects contributing to the temporal organisation of brain activity. By identifying causal, directed interactions between brain regions, this work contributes a step forward in understanding the mechanisms that underlie complex cognitive processes.

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CHAPTER 1

Introduction

Recent research in neuroscience is increasingly shifting from purely physiological descriptions of brain function towards modelling the brain as complex dynamical system (Telesford et al., 2011). This paradigm enables the use of powerful analytical frameworks developed in network science to characterise the structure and dynamics of neural interactions.

1.1 Complex Systems

Complex systems pervade the natural and engineered world, encompassing phenomena as diverse as the coordinated movement of animal flocks and the fluctuations of electrical power grids. Such systems can be represented as large networks of interacting components whose collective behaviour cannot be reduced to the properties of individual elements. A defining feature of these systems is their dynamic nature—the state or activity of each unit evolves over time through interactions with others. These interactions give rise to emergent phenomena, including synchronisation, resilience, and self-organisation. Understanding how such properties arise, and how they may be modulated or controlled, remains a central challenge in modern science with implications spanning neuroscience, physics, and engineering.

To conceptualise the brain as a complex system, it can be represented as a network model in which individual functional units, ranging from single neurons to large scale cortical region, serve as nodes connected via edges (Bassett et al., 2018). The activity at a node propagates across the network according to the connections. The combination of the entire network leads to emergence of the complex dynamics that make these systems interesting. The goal of neural models is to reproduce the emergent phenomena observed in real systems, including oscillatory activity (Khanjaniapak et al., 2024), task-oriented functional organisation (Forrester et al., 2020), and the multistability of coexisting operational regimes (Khanjaniapak et al., 2024).

In neural networks, the topology of connections strongly constrains and shapes the range of dynamics the system can display (Shen et al., 2015). Theoretical studies have shown that indirect interactions between nodes can produce variable patterns of correlation, while highly connected hubs tend to show stronger average correlations (Pernice et al., 2011). To understand the structure of the brain's network is therefore key to begin to explain the mechanisms that give rise to brain function and behaviour. Mapping the brain's connectivity is not only for understanding normal brain function but also for revealing how its disruption contributes to cognitive and neurological disorders.

1.2 Network Inference

It is clear that a method for determining a network's topology would be highly beneficial, as it would allow assessment of the underlying structure (e.g. hierarchical, feed-forward, or recurrent) and thereby constrain and predict emergent dynamics.

However, for many dynamical systems, direct observation of all constituent connections is typically unfeasible. For instance, the precise interactions among the numerous components of protein and gene regulatory networks remain largely hidden within the vast combinatorial space of possible molecular interactions (Timme and Casadiego, 2014). The same limitation applies to the brain. Although techniques such as electrical stimulation exist, where pairs of cortical regions are probed and reciprocal responses to stimulation are used to infer direct or indirect connections. These approaches while providing strong causal evidence at local scales, fail to scale to the resolution and complexity of the entire brain (Jouhanneau and Poulet, 2019). Thus, a more indirect approach that is capable of inferring connectivity from easily observed dynamics becomes essential.

This limitation has motivated a shift from direct probing toward statistical and model-based inference methods. With advances in large-scale recording techniques such as fMRI and EEG, it has become increasingly possible to measure the collective activity of over large regions of the brain with improved spatial and temporal precision.

This shift enables researchers to move from asking "How does structure shape dynamics?" to the inverse question:

"Given observed dynamics, what underlying structure could give rise to them?"

Consequently, there has been a rapid development of methods addressing this inverse problem—inferring network structure from observed neural dynamics (Timme and Casadiego, 2014).

1.3 Contributions

Alongside this development in Computational Neuroscience is have sought to make several major contributions to the field:

- **Methodological Evaluation:** Assessed the performance of linear least squares (LLS) regression for reconstructing directed connectivity from synthetic networks designed to emulate resting-state fMRI dynamics.
- **Group-Level Modelling:** Investigated strategies for constructing representative group effective connectivities, comparing fixed and random-effects approaches to evaluate interpretative differences and robustness.
- **Empirical Application:** Applied LLS inference to HCP resting-state fMRI data to produce a whole-brain effective connectivity, identifying major directional components and network-level patterns.
- **Delay Modelling:** Extended the analysis by fitting a delay-based linear model to quantify how static and derivative terms explain variance in brain activity

1.4 Repository

The entirety of the code the reproduce the work shown throughout this report is at the GitHub repository: https://github.com/BroasaurBot/Brain_Network_Inference_Honours_2025

Chapter 2 begins by outlining the large scale brain structures and activity relating; it back to a description of brain connectivity suitable for network inference.

CHAPTER 2

Background

Understanding how brain structure gives rise to function requires a foundation in both its anatomical organisation and the measurable dynamics that emerge from it. This chapter outlines the relevant biological and methodological background necessary to interpret the subsequent analyses. It begins with an overview of the brain's large-scale structure and the network principles that arise from it, followed by a discussion of how neural activity can be recorded through techniques such as EEG and fMRI. The finally an introduction to the principal descriptions of connectivity: structural, functional, and effective. Highlighting how each description provides a complementary perspective on the organisation and causal interactions of the brain's complex networks.

2.1 Structure and Dynamics

The human brain is a highly sophisticated biological computer, comprised of a macro-scale anatomical structure which supports an underlying complex network of reactive neuronal components.

The brain is hierarchically organised, identified to contain several distinct anatomical regions which are interact through the structural connections. Of all the regions, of particular interest is the cerebral cortex, the outermost layer that envelopes most of the other brain structures, making up approximately 80 percent of the brain volume. The cortex is understood to be responsible for what is called "brainwork" : cognition, visual processing, understanding language and attention (Ackerman, 1992).

The architecture of the brain gives rise to several characteristic network properties that support efficient information processing. At the macroscopic scale, the human connectome exhibits small-world topology, dense local clustering combined with short global path lengths, which enables both specialised and integrated processing (Bassett et al., 2018; Bullmore and Sporns, 2009).

The brain's activity then arises from the coordinated electrical signalling of neurons. These complex networks described shape the signals to generate rhythmic oscillation that propagate locally or distribute throughout the brain, combining to process information and direct responses. At a large-scale this activity appears as fluctuations in the cortical regions and can be captured and measured.

2.2 Brain Activity Recordings

The main measures of brain activity that are essential in contemporary research of network inference are both EEG and fMRI (Omidvarnia et al., 2014; Li et al., 2019; Gilson et al., 2018, 2016). Electroencephalogram (EEG) is a non-invasive technique where electrodes placed on the scalp pick-up voltage changes originating from the action potentials of the neurons. EEG has the advantage of having a relatively high temporal resolution, sampling at 100-1000 Hz, capable of identifying critical frequencies within the brain (Abbas et al., 2021). Although EEG faces several limitations, being sensitive to electrode positioning, signal noise and being unable to directly relate activity to neuronal populations beneath the scalp (Michel and Brunet, 2019). Functional Magnetic Resonance Imaging (fMRI) on the other hand provides high resolution spatial data, obtaining activity measures over the space of 3-4 mm at the trade-off of low temporal resolution, sampling at 1 - 1.5 Hz (Glover, 2011). Critically fMRI supplies recordings over the entirety of the brain capturing hundreds of regions throughout the cortex, this reduces the number of hidden variable or external influences which would otherwise confound the inference between measured regions. These 'whole brain' studies have become increasingly popular with the publication of publicly available fMRI datasets.

2.3 fMRI studies

In actual fact fMRI infers brain activity through detection of the changes in blood oxygenation concentration, BOLD (Blood Oxygen Level Dependent). A cognitive load, whether induced by a task or an unregulated process causes consumption of oxygen in adjacent tissue, altering the magnetic properties of the blood, this change is detected and interpreted as a BOLD signal (Glover, 2011). Due to the slow (~ 5 s-6s) and elongated (~ 3 s) haemodynamic response of blood to neural stimulus the signal requires deconvolution to reproduce a sharp temporal resolution of roughly 100ms (Uruñuela et al., 2024).

fMRI studies typically fall into two categories, rest or task. Task studies record the brain activity of a participant whilst engaged in a 'task', be that speaking or watching a movie. The aim is to investigate

both the temporal dynamics of connectivity and speculate the functional purposes of brain regions. Gilson et al. (2018) investigated how brain areas interact and integrate information when exposed to a stimulus by constructing a temporal dependent effective conductivities, finding that there is increased inter-hemisphere communication during movie viewing.

fMRI may also be applied to a patient at rest, that is the participant is awake, but not engaged in a specific cognitive, motor or sensory task. The hypothesis is that without a cognitive load the the patients rest activity represents background/default activity. This resting activity is thought to be characterised by random uncorrelated noise and asymptotic oscillations around a stable state (Robinson et al., 2014), which improve reliability of statistical analysis. Hence the resting state brain activity regime is impartial to any functional pattern associated with a cognitive task, producing unbiased results. To add, due practicalities rest-studies contain longer time series and more observations which allow for more reliable and faithful EC inference. However there are claims that resting state activity is non-random and exhibit slow fluctuations in spatiotemporal patterns of BOLD signals (Deco et al., 2011). Ultimately rest based studies provide the level of reliability and data richness that make inferencing practical.

2.3.1 fMRI Dataset

The dataset to be examined is The Human Connectome Project (HCP) (Van Essen et al., 2012). It is a large-scale initiative that publicly releases high-quality healthy-brain data. Its resting-state fMRI (rsfMRI) dataset includes 471 subjects scanned at a 720 ms sampling interval ($f_s = 1/0.72 \approx 1.4$ Hz) for approximately 15 minutes, yielding roughly 1,200 samples per subject. Participants were scanned at "rest" (i.e., no overt task), supporting the assumption of stationarity over the recording window. The increased size of this dataset in comparison to earlier sets, allows for larger networks to be reconstructed and more flexible techniques to be applied.

2.4 Connectivity Types

In studying complex systems such as the brain, different philosophies exist regarding the type of network model that best captures the underlying system dynamics. Each framework—structural, functional, and effective connectivity—reflects a distinct interpretation of what it means for two elements to be “connected” and thus supports different inferential goals.

Structural connectivity (SC) refers to the physical connections between neurons or brain regions, often estimated using diffusion MRI or tract-tracing techniques. Although one might assume that a structural link implies strong functional interaction – meaning similarities in dynamics – this is not always the case. Indirect connections and network-wide fluctuations can obscure direct anatomical influences (Timme and Casadiego, 2014). In contrast, functional connectivity (FC) describes the statistical dependencies between the dynamics or activity of pairs of neurons or brain regions, which may even occur between spatially distant regions. However, FC captures only correlations and cannot distinguish whether these dependencies reflect genuine causal influences between neurons (Gilson et al., 2018, 2016). Effective connectivity (EC) seeks to identify directed, causal interactions underlying neural dynamics, typically through model-based approaches such as dynamic causal modelling or Granger causality. It aims to infer the minimal set of directed connections required to reproduce observed activity patterns, and thus necessarily requires multivariate modelling rather than examining pairwise relationships alone. An EC may approach the underlying SC given sufficient data to explore and observe all interactions. In reality observations may be incomplete or lack hidden variables which contribute confounding information, obscuring the SC. Together, SC, FC, and EC provide complementary levels of description—from anatomical structure, to statistical association, to causal influence—and their integration offers a critical framework for understanding how neural networks generate complex, emergent dynamics (Friston, 2011).

With a foundational understanding of the brain’s large-scale organisation and network structure, as well as the terminology used to describe these systems, the next chapter will examine the mathematical basis of existing network inference techniques and their application to fMRI observations.

Literature Review

This chapter will begin with providing an overview on the major design decisions available when developing an inference methods, including: linear/nonlinear neurons, model-based/model-free representations and discrete/non discrete dynamics. It will then examine the current state of resting-state fMRI research, highlighting advancements in functional and effective connectivity as reported in the literature.

3.1 Network Inference Techniques

The promise of inferring the structure of a brain's connections from dynamics has garnered attention from the disciplines of Neuroscience, Physics, Computer science and Biology. Each discipline approaches this 'inverse problem' by making assumptions of the network's model and its applications. The many proposed solutions provide a litany of options to analyse for their assumptions, conditions and accuracy.

3.1.1 Nonlinear vs Linear

A central consideration in network inference is whether to model interactions as fundamentally non-linear, focusing on biological reality or to adopt linear approximations that prioritise tractability and scalability.

3.1.1.1 Nonlinear Microscopic View

Focusing on neuroscience, it has been experimentally established that physical dynamics of individual neurons are inherently non-linear (Nozari et al., 2021). Neurons exhibit non-linear behaviour such as intrinsic refractoriness, where neuron activation drops regardless of excitation, burst firing, where neurons continue to fire without external excitation (Magrans de Abril et al., 2018). Consequently, it would seem

logical for a biologically realistic model of a network's interactions to incorporate nonlinear functions governing self-activation and neuron coupling. Despite their potential for greater biological realism, non-linear models face several disadvantages. They are generally more mathematically complex, requiring sophisticated analytical solutions. The greater computational costs can become preventative to investigate larger networks. They are less data efficient in parameter fitting, requiring vast amounts of data to avoid over fitting.

3.1.1.2 Linear Macroscopic View

At the macroscopic scale, the relationships between large brain regions can often be approximated using linear models, which assume that the activity of each region is a proportional and additive combination of the activity in other regions. This formulation underlies many established approaches in resting-state fMRI analysis, including Pearson correlation, general linear models (GLMs), and multivariate autoregressive (MVAR) models. Linear models gained prominence due to their computational efficiency, interpretability, and analytical tractability, making them suitable for large-scale datasets where model complexity must be balanced against data limitations.

Empirically, several studies have shown that linear models capture a substantial portion of the brain's statistical dependencies. For example, early work demonstrated that simple correlation could recover major resting-state networks, such as the default mode network, accounting for up to ~95% of the mutual information between regional time series (Greicius et al., 2003). More recent analyses have further supported their validity: Nozari et al. (2024) found that, at the whole-brain level, linear models often *outperform nonlinear approaches*, providing superior predictive power at lower computational cost. The authors argue that microscopic nonlinearities are effectively masked at the macroscopic scale due to spatial and temporal averaging, measurement noise, and limited sampling—factors intrinsic to fMRI recordings. Consequently, linear models remain a practically and empirically justified framework for inferring large-scale effective connectivity.

3.1.2 Model Free vs Model Based

Another foundational choice concerns how much structure to assume about the data-generating process. Model-free methods infer relationships directly from observed signals with minimal prior assumptions, while Model-based methods posit an explicit generative mechanism (dynamics and coupling). This

section outlines the strengths and trade-offs of each approach, and why the choice matters for robustness and inference.

3.1.2.1 Model Free

Model-free approaches do not make assumptions about the mechanism that generates the observed data, deducing the connections directly from the patterns in dynamics (Magrans de Abril et al., 2018). Model-free methods are suited to deal with complex interactions where an explicit relationship is difficult to quantify, thus they are suited to describing the non-linear interactions of neurons.

There are several effective applications of model-free methods to infer brain structure. An early application was cross correlation models, they derive causal indicators between neurons from the statistical probabilities (e.g. given that neuron i spikes, what is the probability that neuron j then spikes) (Magrans de Abril et al., 2018). Another example is an information-theoretic approach such as transfer entropy, that describes statistical dependence of neurons. The transfer entropy methods produce directed effective structures from multivariate data with promising results (Novelli et al., 2019; Novelli and Lizier, 2021). A new area of research is the application of deep learning and convolutional neural networks (CNNs) to infer arbitrary relationships between pairs of neurons (Zhang et al., 2019).

While promising, these model-free approaches suffer from common limitations such as high computational demand and the reliance on great quantities of high quality data. As described later in the report, data quantity and quality is often the bottleneck during application and this is particularly true for measures of brain activity (fMRI).

3.1.2.2 Model Based

Model based methods make explicit assumptions of the generative model of network activity, i.e. the mathematical form of the coupling functions and optionally the intrinsic behaviour of each neuron. The aim of a model-based approach is to infer a network that can reproduce the observed data accurately. An advantage of model based methods is once a network is reconstructed, it can generate activity for further comparison and experimentation. There are many existing model based methods in current literature being applied specifically to neural data: popular method includes Autoregressive Models (AR), (Harrison et al., 2003), Generalized linear models (GLM) (Song et al., 2013) and Dynamic Bayesian networks (Eldawlatly et al., 2010). Linear multivariate AR models are currently the most popular of the

model-based inference technique due to their interpretability and low computational cost. To obtain a mathematically tractable model that can scale to the complexity of the whole brain, both physiological and computational simplifications are required, a necessary trade-off for the feasibility of brain simulation (Magrans de Abril et al., 2018).

3.1.3 Discrete vs Continuous time dynamics

Finally, connectivity inference depends on how time is represented. Continuous-time formulations mirror underlying analog networks more closely and connect naturally to differential-equation models, whereas discrete-time formulations align with sampled data (e.g., fMRI TR) and admit convenient estimators like VAR/AR models.

3.1.3.1 Discrete time dynamics

The underlying neural activity can be modelled as a discrete-time process, where the entire network updates synchronously at a regular interval. It is a widely used approach and has the advantages of aligning with the inherently discrete data samples collected from EEGs or fMRIs. An example of a discrete time process is the vector autoregressive (VAR) model (equation 3.1) (Lizier et al., 2023). Equation 3.2 describes a general linearly-coupled system, comprised of N nodes whose activity at time t is a $1 \times N$ row-vector $\mathbf{x}(t) = \{x_1(t), x_2(t), \dots, x_N(t)\}$.

$$x_i(t+1) = x_i(t) + \sum_{j \neq i}^N C_{ji} x_j(t) + \sigma r_i(t) \quad (3.1)$$

$$\mathbf{x}(t+1) = \mathbf{x}(t)\mathbf{C} + \sigma \mathbf{r}(t) \quad (3.2)$$

The connections are described by the adjacency matrix, $\mathbf{C}$ such that each element $C_{i,j} \in \mathbb{R}$ is the strength of coupling from node i to j . In this process the network is driven by mean-zero unit-variance Gaussian noise $\mathbf{r}(t)$, scaled by a stationary noise strength σ .

The aim of network inference would then be given observed data, $\mathbf{G} \in \mathbb{R}^{M \times N}$, where M is the number of time-series samples to reconstruct an effective connectivity (EC) of $\mathbf{C}$, which best describes the activity.

3.1.3.2 Linear Least-Squares

Presented now is the network inference technique that will be investigated throughout the report

An acute observation made by several studies is in the case that $M > N$, there exists for each node independently (3.1) an over-determined linear system of equations. This provides the opportunity to apply linear least squares to analytically solve for $\mathbf{C}$.

Linear least-squares (LLS) is a useful tool to infer the linear relationship between a set of unknown independent variables and known dependent variables (Timme and Casadiego, 2014). The relationship between the sets is assumed to have the form:

$$\mathbf{y} = \mathbf{x}\beta + \epsilon$$

Where $\beta \in \mathbb{R}^{N \times 1}$, is the set of unknown linear coefficients, $\mathbf{y} \in \mathbb{R}^{M \times 1}$ is the set of dependent variables, $\mathbf{x} \in \mathbb{R}^{M \times N}$ is the set of N-independent variables and $\epsilon \in \mathbb{R}^{M \times 1}$ are the random errors (assumed to be independent random variables). The analytical solution of the best estimate $\hat{\beta}$ of the unknown linear coefficients is then given by (3.3):

$$\hat{\beta} = \mathbf{x}^T(\mathbf{x}\mathbf{x}^T)^{-1}\mathbf{y} \quad (3.3)$$

A paper from Shandilya and Timme (2011) adapts the LLS regression technique to derive an analytical method to infer the effective structure of a network, given any general time-series dynamics. The method samples the time series at discrete intervals to approximate the derivative $\dot{\mathbf{x}}_i(t)$, deriving the expression:

$$\dot{\mathbf{x}}_{i,m} = \mathbf{f}_{i,m}(\mathbf{x}_i) + \sum_{j=1}^N \mathbf{C}_{ij}\mathbf{x}_{j,m} \quad (3.4)$$

Where $\mathbf{f}_{i,m}$ is the function of intrinsic coupling (possibly representative of Gaussian noise). Equation (3.4) can then be written in terms of $X_i = \dot{\mathbf{x}}_{i,m} - \mathbf{f}_{i,m}$ which is $\in \mathbb{R}^{1 \times M}$, $\mathbf{C}_i \in \mathbb{R}^{1 \times N}$ and $\mathbf{G}_i \in \mathbb{R}^{N \times M}$ (matrix of M observations), as shown in equation (3.5). Finally this can be rewritten in the form of equation (3.3):

$$\mathbf{X}_i = \mathbf{C}_i\mathbf{G}_i \quad (3.5)$$

$$\hat{\mathbf{C}}_i = \mathbf{X}\mathbf{G}_i^T(\mathbf{G}_i\mathbf{G}_i^T)^{-1} \quad (3.6)$$

Solving this for all $i \in 1, \dots, N$ yields the complete structure of the network, $\hat{\mathbf{C}}$. Note for later that $\mathbf{G}_i \mathbf{G}_i^T$ is also known as the covariance matrix $\mathbf{\Omega}$, where $\mathbf{\Omega}_{ij}$ is the time averaged covariance of variables $\mathbf{x}_i$ and $\mathbf{x}_j$.

Pik-Yin Lai proposes a related approach for inferring directed network structures from discrete-time series data in systems influenced by white noise (Lai, 2017). The paper begins with a baseline equation similar to (3.1) and then continues to only consider small deviations $y_i(t) = x_i(t) - X_i(n)$ from the time-independent noise free solution X_i . It rewrites equation (3.1) as

$$\mathbf{y}(t+1) \approx \mathbf{N}\mathbf{y}(n) + \sigma \mathbf{r}(t) \quad (3.7)$$

Where $\mathbf{N}$ has adsorbed both the weighted connectivity matrix and the first order derivatives of the coupling functions. The time-lagged correlation, $\mathbf{\Omega}_{\tau+1}$ can then be calculated by:

$$\mathbf{\Omega}_{\tau+1} = \langle \mathbf{y}(t+\tau+1) \mathbf{y}^T(t) \rangle \quad (3.8)$$

$$= \langle \mathbf{N}\mathbf{y}(t+\tau+1) \mathbf{y}^T(t) \rangle + \sigma \langle \mathbf{r}(t+\tau+1) \mathbf{y}^T(t) \rangle \quad (3.9)$$

$$= \mathbf{N}\mathbf{\Omega}_t \quad (3.10)$$

Where $\langle \mathbf{r}(t+\tau+1) \mathbf{y}^T(t) \rangle = 0$, under the assumption that white noise is mean centred and independent of activity, averaging out to zero over long enough observation periods. This leads to another analytical solution for the representative connectivity $\mathbf{N}$:

$$\mathbf{N} = \mathbf{\Omega}_{\tau+1} \mathbf{\Omega}_\tau^{-1} \quad (3.11)$$

While the derivations of both papers appear dissimilar, once you consider that $\mathbf{\Omega}_i = \mathbf{G}_i \mathbf{G}_i^T$ the solutions equate (other than the terms absorbed in $\mathbf{N}$ (Lai, 2017)), suggesting that both methods rely on the analytical solution of linear least squares.

Both studies then appraise effectiveness of their reconstructions by applying their methods on synthetic networks. The paper from Shandilya and Timme (2011) claims their reconstruction technique applies to any general dynamical system, therefore it is demonstrated on four distinct Rössler oscillators that range from simple periodic synchronous dynamics to very complex, highly chaotic asynchronous states. Lai (2017) applies their technique to Erdős-Rényi random networks and a variety of coupling functions, connection distributions and network sizes. Both methods report high quality reconstructions of $\mathbf{C}$, with error rates below 0.1%. Notably a useful metric to demonstrate the quality of the reconstruction, was

both the false negative and false positive rates, as they indicate the methods biases to produce sparse or over fitted networks.

Shandilya and Timme (2011) demonstrates sub-linear scaling of observation time vs the network size and the ability to regenerate networks with up to $N > 1000$ nodes. This indicates the LLS regression is capable of reconstructing large networks on the scale of the whole brain ~ 300 nodes.

An inherit assumption of the LLS regression is the existence of a valid solution $\hat{\beta}$ (equation 3.3), which may not hold in the presence of noise. Both studies test the robustness of their methods under significant noise, Lai (2017) reports that their solution is "insensitive to noises of different strengths and distributions", while Shandilya and Timme (2011) demonstrates successful network reconstruction even under "substantial" noise.

Under the assumptions of linear coupling and discrete-time dynamics, linear least squares regression offers an analytically efficient approach for inferring a network's effective structure from time-series data. The method reliably reconstructs large-scale networks, even under substantial noise, demonstrating its applicability to whole-brain analyses and real-world neuroimaging data.

3.1.4 Continuous time dynamics

The network dynamics can alternatively be represented as a continuous process, modelling activity using differential equations. The continuous model is widely used, as it is considered more realistic for modelling real biological processes as apposed to the discrete model (Lizier et al., 2023). An example process is the multivariate Ornstein-Uhlenbeck model.

$$d\mathbf{x}_i(t) = \left[-\mathbf{x}_i\theta_i + \sum_{j \neq i}^N \mathbf{x}_j(t)C_{ji} \right] dt + \sigma_i d\mathbf{W}_i(t) \quad (3.12)$$

$$d\mathbf{x}(t) = -\mathbf{x}(t) \cdot (I - \mathbf{C})\theta dt + \sigma d\mathbf{W}(t). \quad (3.13)$$

Similar to (3.1), $\mathbf{x}(t)$ is the activity of the N nodes at time t , $\mathbf{C} \in \mathbb{R}^{N \times N}$ is the connectivity matrix and σ^2 is the noise strength. Newly introduced is reversion rate θ and the multivariate Wiener process $\mathbf{W}.(t)$ (Barnett et al., 2009).

Several studies (Gilson et al., 2016, 2018; Steuer et al., 2003; Barnett et al., 2009) derive relationships linking the connectivity to time-lagged covariances in the continuous regime, under the assumption of stationarity (i.e the network fluctuates near a stable state) similar to the discrete formulation of (3.11).

Once such formulation comes from Gilson et al. (2016) :

$$-\Sigma = \mathbf{J}\Omega_0 + \Omega_0\mathbf{J}^T \quad (3.14)$$

$$\Omega_\tau = \Omega_0 \cdot \expm[\mathbf{J}^T \tau] \quad (3.15)$$

Where the Jacobian $J_{ij} = -\delta_{ij}\theta_i + \mathbf{C}_{ij}$ is the combination of the connectivity and reversion rate, Σ is a noise matrix with terms $\sigma_i^2 = \delta_{ij}\langle d\mathbf{W}_i(t)d\mathbf{W}_j(t) \rangle$ (non-diagonal entries are zero because white noise is uncorrelated by construction) and $\expm$ is the matrix exponential. It might appear attractive to derive a solution for $\mathbf{J}$ from equations (3.14) and (3.15), using the covariance matrix. Although (Galán, 2008; Ren et al., 2010) raise the issue that due to the asymmetry of $\mathbf{J}$, it has a twofold degree of freedom compared to symmetric Ω_0 , and that solving for $\mathbf{J}$ in (3.14) results in a nonlinear equation with no unique solution. If comparing (3.15) and relating it to (3.10), it is clear that retrieving $\mathbf{J}$ is not as simple as multiplying by Ω_0^{-1} . The mathematics of the continuous process prove to be tractably harder to derive an analytical relationship and require further treatment.

3.1.4.1 Continuous Solution

Early approaches addressed the problem analytically by assuming a symmetric or bidirectional network structure, which simplifies the Lyapunov equation linking the covariance and connectivity matrices and enables direct inversion for $\mathbf{C}$ (Ren et al., 2010; Ching et al., 2013). However, this symmetry assumption limits applicability to realistic, directed brain networks. More general methods therefore adopt optimisation-based solutions, such as Lyapunov optimisation (Gilson et al., 2016), which iteratively adjusts $\mathbf{C}$ to minimise the mismatch between model-predicted and empirical time-lagged covariances. This approach effectively constrains an otherwise underdetermined system and has been successfully applied to both simulated and empirical fMRI data. Alternatively, compressive-sensing frameworks (Wang et al., 2011) reformulate the problem as a sparse regression, recovering $\mathbf{C}$ by minimising the L_1 norm under dynamical constraints, allowing accurate reconstruction from limited measurements when the network is assumed sparse.

While continuous-time models offer greater biological realism, solving for the underlying connectivity often requires complex or restrictive assumptions, ultimately reducing the interpretability and practical applicability of the results.

3.1.5 Technique Comparison

The approaches reviewed above can be grouped along three overarching design axes: (i) linear vs. nonlinear interactions, (ii) model-based vs. model-free inference, and (iii) discrete- vs. continuous-time dynamics. Each axis reflects a trade-off between biological descriptiveness (how closely the model mirrors neural mechanisms) and practical considerations (interpretability, computational efficiency, and mathematical tractability).

3.1.5.1 Trade-offs in brief

- **Descriptiveness vs. tractability:** Nonlinear, continuous, and model-free methods are often more expressive and can capture rich neural phenomena, but they typically require more data, heavier computation, and bespoke estimation procedures. Linear, discrete, and model-based methods sacrifice some biological detail in exchange for closed-form estimators, clearer identifiability conditions, and scalable computation.
- **Interpretability vs. flexibility:** Model-based formulations yield parameters with direct mechanistic meaning (e.g., coupling weights, time constants), aiding hypothesis testing and simulation. Model-free approaches offer flexibility and fewer assumptions, but parameters are less tied to physiology and results may be harder to explain mechanistically.
- **Data demands and robustness:** High-capacity nonlinear or model-free estimators can overfit when data are limited or noisy (common in fMRI). Simpler linear, discrete, model-based estimators are typically more data-efficient and come with well-understood statistical guarantees.

3.1.5.2 At-scale inference on fMRI

For whole-brain fMRI (hundreds of regions, thousands of observations), **linear, discrete-time, model-based** techniques (e.g., VAR/AR families with least-squares or regularised estimators) provide an attractive operating point:

- (1) They admit *analytical or closed-form solutions* (e.g., least squares), enabling efficient estimation and principled uncertainty assessment.
- (2) They are *interpretable*: coefficients directly encode directed influence under the model, and the framework supports simulation and counterfactual testing.

- (3) They *scale* to brain-level graphs without assuming specific nonlinearities or requiring specialised optimisation.

While these choices may under-represent nonlinear or state-dependent effects, they deliver reproducible, computationally feasible baselines.

3.2 Brain Functional Connectivity

As indicated earlier, there is a wealth of existing research into the functional connectivity of the brain, capturing the statistical relationships between regional activities and relating them to cognitive and physiological processes. Of particular importance to this study is the development of parcellation schemes, which subdivide the cortex into spatially distinct yet functionally coherent regions or “parcels.” These schemes provide the fundamental unit of analysis for network inference, determining how brain activity is aggregated and how connections are defined.

3.2.1 Gordon’s Parcellation

The work by Gordon et al. (2016) introduced a cortical parcellation derived from resting-state functional connectivity boundary maps, aiming to define parcels that correspond to meaningful cortical regions. Their method involved identifying abrupt transitions in voxel-wise connectivity patterns (i.e., boundary maps) and then creating parcels that exhibited high internal homogeneity in resting-state connectivity. They showed that these parcels broadly aligned with known architectonic areas and that the resulting parcellation was more homogeneous than null models or alternative schemes. The purpose was to generate a functionally meaningful “node” set for network-based analyses of fMRI and connectivity. The parcellation has since been widely adopted (e.g., as the 333-parcel Gordon atlas) and has facilitated more fine-grained and reproducible network analyses.

3.2.2 Yeo’s Functional Modules

The seminal study by Yeo et al. (2011) mapped the organisation of the human cerebral cortex via resting-state functional connectivity in 1000 subjects, and used clustering to derive large-scale network modules (initially 7- and 17-network solutions), a more coarse-grain solution than Gordon’s parcellations. The aim was to identify a set of brain regions which exhibit reliable homogeneous functional activity which

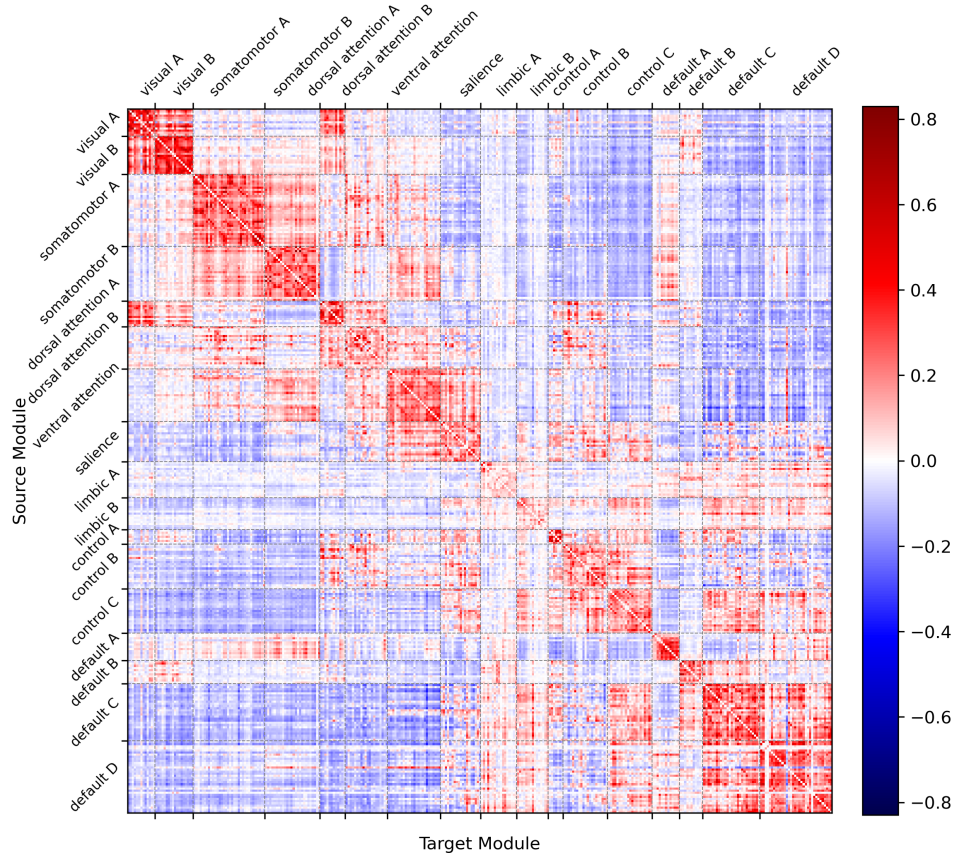


FIGURE 3.1. **Average Functional Connectivity of Human Brain.**

Matrix describes the region-region time average correlation, average across all subjects of the rs-fMRI dataset according to equation 4.5) Regions ordered and labelled according to Yeo's functional modules.

may be related to cognitive functions. A demonstration of the module correlation is in Figure 3.1, calculated in subsequent chapter 5. Their approach showed that sensory/motor networks are organised locally and topographically, while association networks (default, fronto-parietal, dorsal/ventral attention) are more distributed. The modules have proven effective: these large-scale networks have been replicated across datasets, used as standard templates for many connectivity studies, and shown to explain substantial portions of variance in resting-state data (e.g., distinct networks for visual, somatomotor, attention, default, control). However, limitations remain: the clustering is group-level (not individual), the resolution is coarse, and some networks may be heterogeneous internally. Future work should aim at refining these modules with individual precision mapping, dynamic (time-varying) network definitions, and integration with structural connectivity and behaviour.

These parcellations were derived from resting state data. To aid interpretation of the connectivity analyses presented later, a brief overview of the principal functional roles associated with each of these cortical modules is provided below.

3.2.3 Functional module descriptions:

- (1) Visual Cortex(VI): responsible for processing visual input (Huang and Wilkins, 2021). Can be further specified in to visual A, responsible for foveal/central vision and visual B, peripheral vision (Zhang et al., 2024).
- (2) Somatomotor (SM): supports the execution and control of body movement, along with somatosensory processing of sensory information. A & B partitions are spatially segregated and correlated to distinct body parts (Yeo et al., 2011).
- (3) Dorsal Attention (DA): Also known as fronto-parietal attention network, supports top-down goal-directed attentional orienting (Corbetta and Shulman, 2002).
- (4) Ventral Attention (VA): supports bottom-up, stimulus driven-attention, detecting unexpected changes in the environment and redirecting attention (Corbetta and Shulman, 2002)
- (5) Salience (S): monitor internal and external salient events and performs a key role in switching between internally directed (default-mode) and externally directed (attention/control) networks (Menon and Uddin, 2010) .
- (6) Limbic (L): is associated with emotional processing, reward/vale and memory encoding (Catani et al., 2013)
- (7) Control (CON): Also known as Central Executive Network (CEN), supports cognitive control, goal directed behaviour and working memory and is most active during cognitive tasks (Mulders et al., 2015)
- (8) Default mode network (DMN): is most during rest and is associated with mind-wandering and internally-directed cognition (Raichle, 2015).

More recent studies have shown that the distinction between the bottom-up and top-down attention networks: Dorsal and Ventral may not be as distinct as once believed and that their cooperation enables the dynamic control of attention (Vossel et al., 2014). Neuroscience is a developing field, theories and models of the brain function grow in popularity and then are later scrutinised.

3.3 Brain Effective Connectivity

This brings the review of inference methods and brain connectivity up to recent developments over past 5 years of brain effective connectivity inference. This section introduces two of the articles, their methods, scope and findings, so that the report may establish grounds for further study and contribution to the field.

"Model-based whole-brain effective connectivity to study distributed cognition in health and disease" from Gilson et al. (2020) builds upon their previous work (Gilson et al., 2018) on a model based multivariate Ornstein–Uhlenbeck (MOU) stochastic process, termed MOU-EC, inference technique. The aim was to investigate how cognitive processes are distributed across brain networks in healthy and diseased states. The MOU-EC technique is a linear model scaffold-ed off prior-knowledge of the structural connectivity. Which is then applied directly to the BOLD signal, modelled as a continuous time multivariate autoregressive process. The technique was demonstrated using task-based fMRI data, where distinct connectivity patterns and network biomarkers were identified that discriminated between cognitive tasks. The results highlight the advantages of linear modelling, enabling reliable reconstruction of directed connectivity from relatively short recordings (approximately 10 minutes at $TR = 2$ s) and allowing for meaningful network interpretation. However, the analysis was limited by its coarse spatial resolution, reconstructing only 66 regions and thus lacking the ability to resolve finer intra-modular connections.

"Sparse DCM for whole-brain effective connectivity from resting-state fMRI data" by Prando et al. (2020) introduces a novel technique building on dynamic causal modelling for estimating whole brain effective connectivity from resting-state fMRI data. The method is again a linear model, that integrates sparse estimation methods. The sparse DCM method was evaluated on both simulated and empirical data. The rs-fMRI data included 24 subjects, extracting 66 cortical regions according to the Hagmann atlas. The sparse DCM technique provided accurate reconstructions of simulated data and importantly, since the model promoted sparsely directed networks, graph-theoretical analyses could be applied to identify nuanced asymmetries not present in FC studies. While the sparse DCM marked a step toward whole-brain effective connectivity, it faced two key limitations. The computationally intensive sparsity constraints, restricted the size of network to 66 regions. The method also required a subjective choice of threshold to allow the models to converge, which meant a trade-off between sensitivity and specificity.

The field of effective connectivity is rapidly advancing, driven by the growing availability of high-quality datasets. In particular, the Human Connectome Project (HCP) resting-state fMRI dataset offers a large number of subjects and extensive temporal coverage, creating new opportunities to apply and evaluate inference techniques such as linear least squares (LLS). This scale enables more robust group-level analyses and the exploration of directed connectivity across the whole brain.

3.4 Literature Summary

The reviewed literature highlights several key insights that guide this investigation:

- (1) Compared to functional or structural connectivity, **effective connectivity (EC)** offers a framework that captures directional and causal relationships between brain regions while remaining experimentally and computationally feasible to estimate.
- (2) A **linear model** of brain connectivity, reconstructed from discrete-time dynamics, can capture meaningful and interpretable patterns of interaction while maintaining mathematical simplicity and computational efficiency.
- (3) **Resting-state fMRI (rs-fMRI)** data exhibit stable, low-frequency fluctuations that are thought to reflect cognitive modes. These properties make rs-fMRI particularly well-suited for whole-brain EC inference.
- (4) To the best of current knowledge, **linear least-squares regression** has not yet been systematically applied to rs-fMRI data to reconstruct a whole-brain directed effective connectivity network.

3.5 Research Questions

These factors above neutrally guide the following research questions

- (1) How accurately can linear least-squares regression recover known connectivity structures from synthetic datasets that mimic fMRI dynamics and sampling limitations?
- (2) When applied to resting-state fMRI recordings, what effective connectivity structure emerges at the whole-brain level, and what are its potential biological interpretations?

The following chapters will express in detail the steps taken to answer these research questions and a discussion on the findings.

Synthetic Data

Introduction

This chapter tests a linear least-squares (LLS) regression estimator on synthetic data generated by a multivariate vector autoregressive (VAR) model, with the goal of recovering the directed, underlying connectivity that governs the VAR dynamics. Using synthetic VAR data where the ground-truth network is constructed has several advantages:

- (1) Control over hidden variables
- (2) Characteristics of the network (connectivity, randomness)
- (3) Propagation speed of activity
- (4) Critical evaluation of the accuracy of the reconstruction

As introduced earlier the aim is to be able to apply the LLS regression to infer the directed EC of the whole brain using the resting-state fMRI (rs-fMRI) dataset produced by the Human Connectome Project (presented section 2.3.1).

Importantly the HCP rs-fMRI dataset demonstrates several key qualities:

- Includes 471 distinct subject recordings
- Scan is parcellated into $N = 333$ voxels (regions) of the cortex, arranged following Gordon's parcellations
- Each scan contains $\approx 1,200$ samples
- Sampled at 720 ms sampling intervals ($TR = 0.72s$), resulting in low-frequency activity
- Participants were scanned during 'rest' supporting the assumption of stationarity over the recording window.

To emulate resting-state fMRI, the synthetic experiments are matched to typical regimes: network size $N \approx 300$, sample length $M \approx 10^3$ (on the order of a single scan), infra-slow dynamics driven by high intrinsic persistence, and observations that are temporally smoothed and *subsampling* to mimic the low-pass effects of the haemodynamic response and finite TR.

4.1 Method Outline

Before proceeding with the investigation, this section outlines the key steps undertaken to develop a robust analysis of linear least-squares. While the steps are presented sequentially for clarity, the development of Chapters 4 and 5 occurred iteratively. Techniques were first developed and validated on synthetic datasets, then applied to resting-state fMRI (rs-fMRI) activity; insights from these analyses informed further refinement of both models and methods.

The synthetic investigation will be presented in the following stages:

- (1) **Synthetic network construction:** Generate synthetic brain networks with small-world properties and simulate corresponding time-series activity representative of neural dynamics.
- (2) **Robustness to network and sample size:** Evaluate the accuracy and stability of LLS inference across networks of varying size and across different sample lengths.
- (3) **fMRI signal emulation:** Introduce infra-slow temporal fluctuations and sampling limitations to synthetic data to mimic the constraints of fMRI acquisition.
- (4) **Inference under fMRI-like conditions:** Assess the robustness and reliability of LLS reconstruction when applied to these fMRI-emulated datasets
- (5) **Analyses and simple aggregation** (“boosting”) strategies that clarify which timescales are recoverable and how group-level estimates

This is all in order to aim the question:

How accurately can linear least-squares regression recover known connectivity structures from synthetic datasets that mimic fMRI dynamics and sampling limitations?

Throughout this chapter, the reports of reliability are the standard deviation of measures generated from 10 trials

4.2 Network Connectivity

To generate sample activity to perform inference on, $\mathbf{X} := \{\mathbf{x}(0), \mathbf{x}(1), \dots, \mathbf{x}(M)\}$ according to the discrete VAR model (3.1), the ground truth underlying connectivity, $\mathbf{C}$ must first be determined. The Watts-Strogatz network (Watts and Strogatz, 1998), are a suitable model for brain connectivity, exhibiting the small-worldness present in neural topology. The particular method used is similar to the original proposed in Watts and Strogatz (1998) paper, although with the addition of directed edges. An outline on the method is as follows:

A regular ring lattice of size N is formed, each node v_i is connected via a directed edge to $\lfloor k/2 \rfloor$ of its left and right neighbours, introducing k edges: $\{v_i \rightarrow v_{i-\lfloor k/2 \rfloor}, \dots, v_i \rightarrow v_{i+\lfloor k/2 \rfloor}\}$. In this sense k/N represents the percentage connectivity of the network; as $k/N \rightarrow 1$, the number of connections scale linearly and the network becomes fully connected. With a probability of p , each edge is randomly redirected to another node. With just a $p > 0.01$, the network reaches both clustering coefficient and network efficiency of that of the brain (Watts and Strogatz, 1998). These combined steps achieve the results shown in Figure 4.1 .

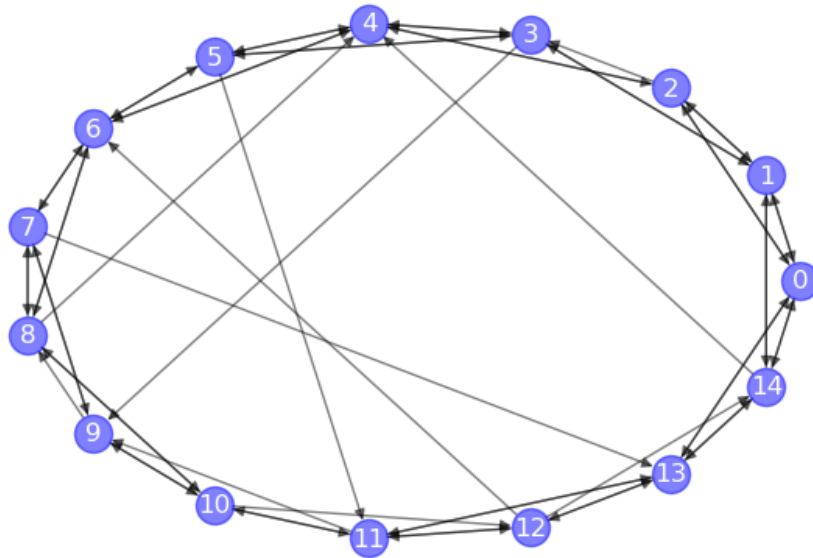


FIGURE 4.1. **Directed Watts-Strogatz network.**
($N = 15$, $k = 4$, $p = 0.1$)

The graph is represented as a weighted, directed adjacency matrix $\mathbf{C}$, where $C_{ij} \in \mathbb{R}$ is the connection strength from node $v_i \rightarrow v_j$. Each connection is assigned a random weight drawn from a uniform distribution, $U[-1, 1]$, allowing for both excitatory, $C_{ij} > 0$ and inhibitory connections $C_{ij} < 0$ connections. Whilst papers such Lai (2017); Ching et al. (2013) allow for general non-linear functions to coordinate the intrinsic activation/self connectivity of a node, this work applies a simpler auto regressive approach (equation 4.3:

$$x_i(t+1) = f(x_i(t)) + \sum_{j \neq i}^N C_{ji} x_j(t) + \sigma r_i(t) \quad (4.1)$$

$$f(x_i(t)) = \alpha x_i(t) \quad 0 \leq \alpha < 1, \alpha \in \mathbb{R} \quad (4.2)$$

$$(4.3)$$

Where α is the intrinsic connectivity, similar for all nodes in the network. The linear intrinsic connectivity can then be absorbed into the connectivity matrix, $\mathbf{C}$ such that each diagonal $C_{ii} = \alpha$.

The VAR model 3.2 is a system of coupled linear equations and therefore the eigenvectors, $\mathbf{v}_i$ and eigenvalues, λ_i of $\mathbf{C}$ are the modes and the respective magnitude of evolution of activity over time. To ensure stability and avoid an explosion of activity, the max eigenvalue or spectral radius is restricted to: $\max |\lambda_i| = \rho(\mathbf{C}) \leq 1$ (stability condition) (Robinson et al., 2014; Gilson et al., 2018; Barnett et al., 2009). This is computed by iteratively updating $\mathbf{C} \rightarrow \mathbf{C}/\rho(\mathbf{C})$, whilst maintaining $C_{ii} = \alpha$. With this the synthetic networks can be constructed as depicted in Figure 4.2.

4.3 Activity

With a synthetic network, the M samples of activity, $\mathbf{X} \in \mathbb{R}^{M \times N}$ are then iteratively generated according to equation 3.2. This is given the initial conditions for each node $x_i \sim \mathcal{N}(0, \sigma^2)$ and a stationary noise strength σ . A sample activity is provided in Figure 4.3, generated from the network demonstrated in Figure 4.2. Since the growth of activity is bounded by the stability condition, the choice of σ dictates the time-average magnitude of activity. This is demonstrated in Figure 4.3, where for these parameters the time-averaged activity $\overline{\langle |\mathbf{x}(t)| \rangle} = 0.29$ in fact corresponds to the setting for $\sigma = 0.3$. For the remainder of the Chapter 4, $\sigma = 0.3$ is used since it distributes activity generally in the range $\mathcal{N}(0, \sigma^2)$.

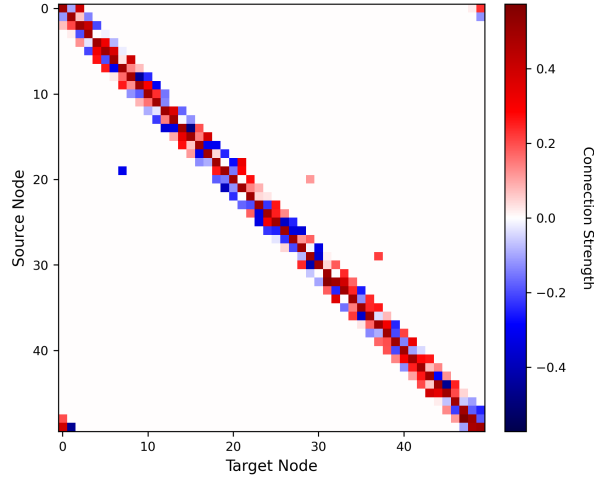


FIGURE 4.2. **Adjacency matrix representation of a network connectivity instance.** The network sample uses the parameters: $N = 50, k = 4, p = 0.05, \alpha = 0.5$. Red indicates excitatory and blue indicates inhibitory connections.

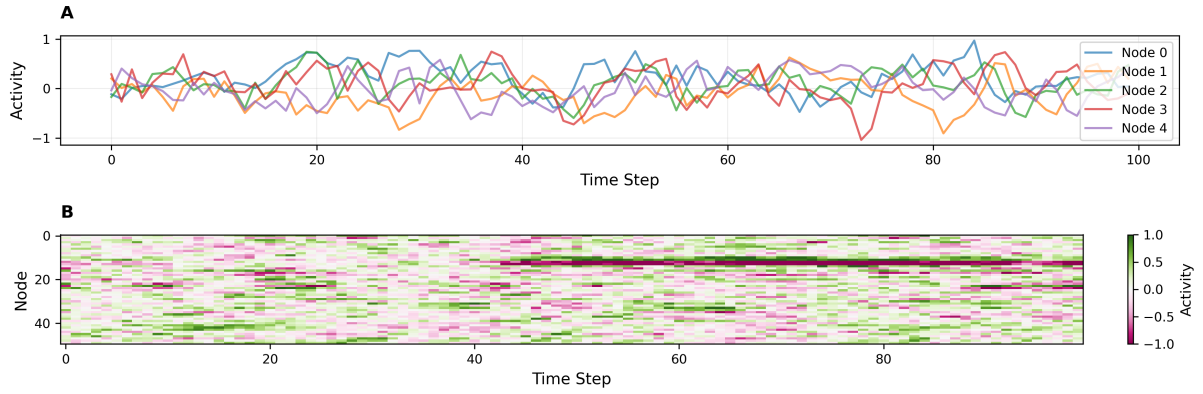


FIGURE 4.3. **Sample Generated activity.** Activity generated from network described in Figure 4.2 and parameter: $\sigma = 0.3, \sigma_i = 0.3$ (A): Activity of sample nodes (0-4) for 100 time steps. (B) Heat map representation of activity of all nodes over the 100 time steps.

4.4 LLS Reconstruction

As stated in the introduction, the aim is to infer the directed-weighted network connectivity, $\hat{\mathbf{C}} \in R^{N \times N}$ from the time series activity, $\mathbf{X} \in R^{M \times N}$. The proposed inference technique follows the work done by Lai (2017), in which applies principles of linear least squares regression (3.3) to solve for matrix weights (equation 3.11). Explicitly, the method to reconstruct $\hat{\mathbf{C}}$ given $\mathbf{X}$ is as follows. First, recall that from the time-series activity we can measure Ω_τ , the averaged time-lagged correlation between activity, $\mathbf{x}$ at time t and $t + \tau$ (noting that the expectation value $\langle \dots \rangle$ is approximated via an ergodic assumption as the

average over all time $t \in (0, M]$).

$$\mathbf{\Omega}_\tau = \langle \mathbf{x}(t + \tau) \mathbf{x}^T(t) \rangle \quad (4.4)$$

$$= \frac{1}{M - \tau} \sum_{t=0}^{M-\tau} \mathbf{x}(t + \tau) \mathbf{x}^T(t) \quad (4.5)$$

Then, following similar steps to equation 3.10, the connectivity simply is inferred as:

$$\hat{\mathbf{C}} = \mathbf{\Omega}_1 \mathbf{\Omega}_0^{-1} \quad (4.6)$$

Due to the assumption of normally distributed Gaussian noise, $\sigma \mathbf{r}(t)$ the LLS regression performs the role of a maximum likelihood estimator. Optimising the parameters, C_{ij} under the assumed the probability distribution, results in the most probable $\hat{\mathbf{C}}$. This technique also assumes that there is sufficient number of samples to satisfy an overdetermined linear system of equations. The performance the technique hinges on this requirement and hence will be a focus of further investigation.

Several parameters have been introduced so far: network size (N), sample size (M), intrinsic connectivity (α). The effect of these parameters on the resulting quality of reconstruction will be further explored to understand the limit of the LLS regression technique.

4.5 Preliminary Demonstrations

4.5.1 Successful Reconstruction

We begin by presenting a demonstration of the reconstruction of a synthetic network in Figure 4.4. With $M = 10^5$ the reconstruction was clearly able to reproduce the characteristic ring network and the swapped edges. To properly access the quality of the reconstruction it's useful to plot the $\hat{C}_{ij}$ against the C_{ij} as shown in (B). The vertical spread of the $\hat{C}_{ij}$ about the $y = x$ trend is an indication of the quality: the closer the points are to $y = x$, the better the quality of the reconstruction. In the figure there are two clear vertical clusters about $C_{ij} = 0$ & $C_{ij} = 0.5$, which are the (large numbers of) non-connections and the self connections respectively.

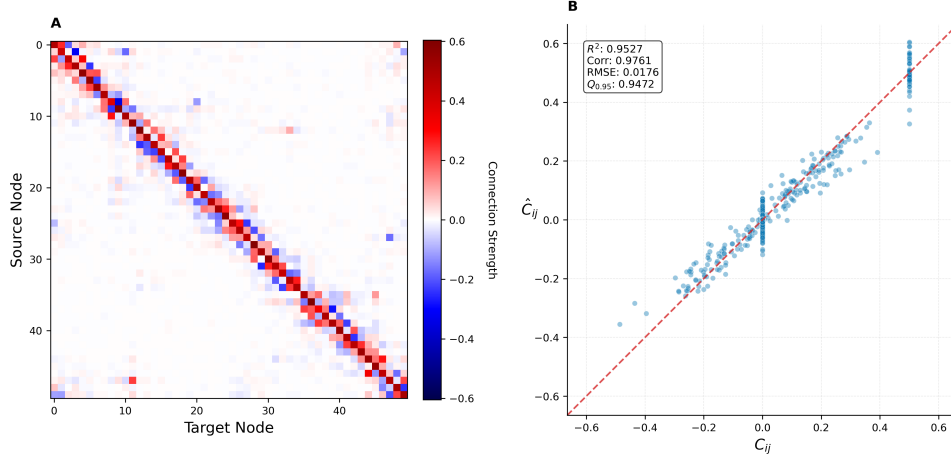


FIGURE 4.4. **Successful reconstruction, $\hat{C}$.**

Network parameters: $N = 100, k = 4, p = 0.05, \alpha = 0.5, \sigma = 0.3$, with a sample size, $M = 10^5$. (A) Connectivity plot of the reconstructed network. (B) Comparison of the reconstructed connections $\hat{C}_{ij}$ against the true connections C_{ij} . Verification includes a R-squared, Correlation, Residual error and $Q_{0.95}$ (a measure of the 'Quality of Reconstruction').

4.5.2 Measurements of Quality

There are several quantitative measures of the performance of the reconstruction technique. The coefficient of determination (R^2) quantifies the proportion ($0 \leq R^2 \leq 1$) of variance in the true connectivity explained by the reconstructed connectivity. The Pearson correlation coefficient (r) similarly assesses the strength and direction of the linear relationship between $\hat{C}_{ij}$ and C_{ij} . The root mean square error (RMSE) measures the standard deviation of the residuals between $\hat{C}_{ij}$ and C_{ij} , representing the average deviation of the reconstructed values from the ground truth, as illustrated in Figure 4.4B. Finally, the quality of reconstruction (Q_q), introduced by Shandilya and Timme (2011), represents the fraction of reconstructed connection weights that fall within $(1 - q)$ of their true values, providing an interpretable measure of reconstruction accuracy.

4.5.3 Unsuccessful Reconstruction

Figure 4.5 is an example of an unsuccessful reconstruction of the connectivity from insufficient number of time samples, $M = 200$. It is clear from both Figure 4.5(A & B) – particularly in comparison to Figure 4.4 – that there is significant error in the reconstruction, with an increase in RMSE: $0.0176 \rightarrow 0.0320$ and a drop in quality, $Q_{0.95} : 0.94 \rightarrow 0.46$. In particular there is an increase in the variance of the

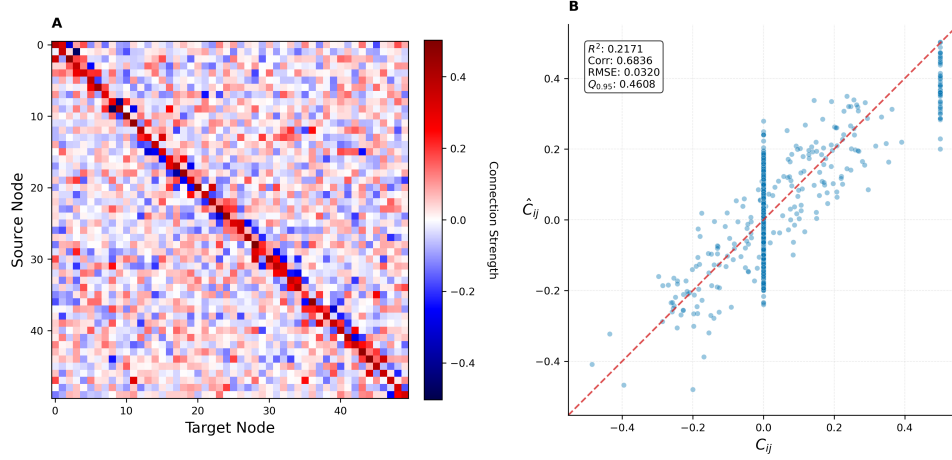


FIGURE 4.5. **Unsuccessful reconstruction of a $\hat{\mathbf{C}}$.** Network same as Figure 4.4: ($N = 100, k = 4, p = 0.05, \alpha = 0.5, \sigma = 0.3$), with fewer samples, $M = 200$.

predicted coupling strengths $\hat{C}_{ij}$ of the non-connections, at $C_{ij} = 0$, with their range almost completely overlapping with the predicted coupling strengths of the truly connected edges.

4.6 Sample Size Requirement

To solve the system of coupled linear equations, there must be more known independent variables than unknown dependent variables (equation 4.6). In this case, it implies: the size of $\mathbf{X}$ is greater than $\mathbf{C} \implies M \times N > N^2 \implies M > N$.

To investigate this several network sizes reconstructed from increasing number of sample observations. Figure 4.6 demonstrates this, as $Q_{0.95} \approx 0$ for any $M < N$ and for values of $M > N$ the quality climbs reaching $Q_{0.95} > 0.95$ in around $3 \times N$ many samples.

Figure 4.6B displays the effect of increasing network size with a fixed number of samples. The result is a similar trend for $N < M/2$ the $Q_{0.95} \approx 0.9$ and drops to $Q_{0.95} \approx 0$ for any $N > M$. This sharp transition indicates the inference technique becomes inaccurate as $N \rightarrow M$.

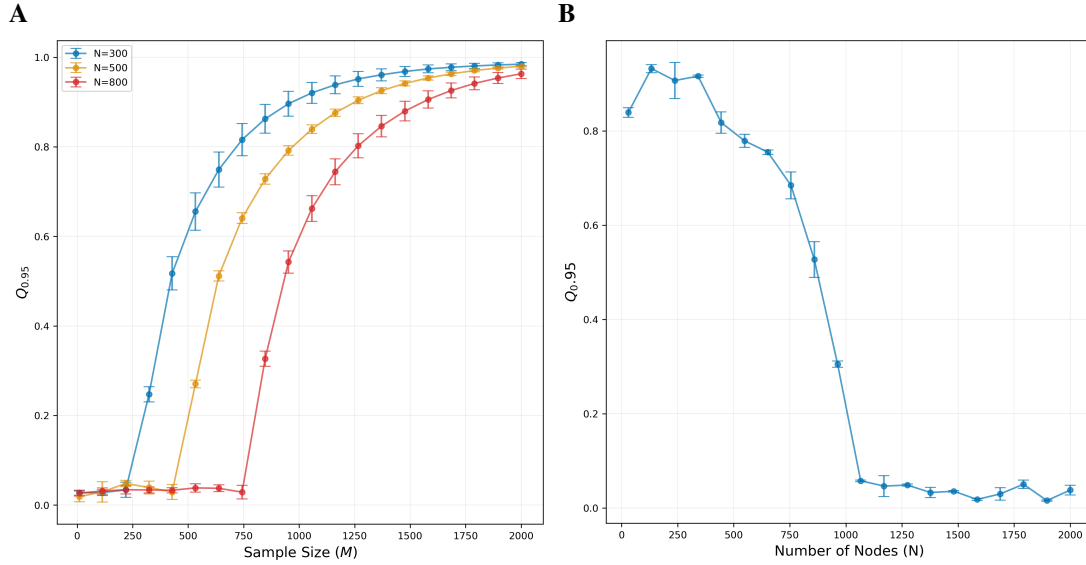


FIGURE 4.6. **Impact of sample & network size on inference quality.** Network parameters: ($p = 0.05, k = 4, \alpha = 0.5, \sigma = 0.3$). (A) $Q_{0.95}$ for increasing M , for three sized networks $N = 300, 500, 800$. (B) $Q_{0.95}$ for increasing sized networks, using $M = 1000$.

4.7 Subsampling

So far the inference technique has been applied to activity directly generated by the underlying network. Though under the circumstances of fMRI recordings the activity is altered by several physiological and statistical processes.

The ability for fMRI to detect the underlying neuronal signals is limited both by the physiological Haemodynamic Response Function (HRF) and the detected BOLD signal. The HRF acts as a low-pass filter, temporally smoothing the underlying activity over ranges of 4-6 seconds (Gomez et al., 2024). Then the measured BOLD is a convolution of neural activity with the HRF, which is theorised to be "proportional to local average neural activity averaged over a period of time," (Boynton et al., 1996).

Therefore to approximate the effect of the practical limitations (both temporal resolution and the HRF) of fMRI, the network inference technique will be applied to activity that has been passed through a 'subsampling & smearing' process.

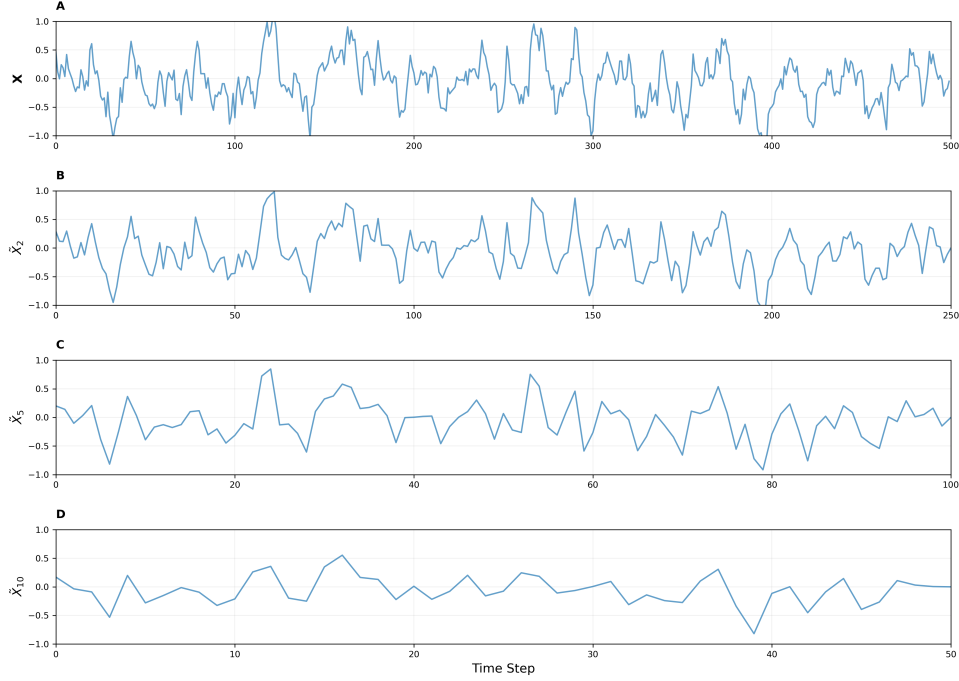


FIGURE 4.7. **Subsampling network activity.**

(A) Original activity, $\mathbf{X}$ is 500 samples long. (B) Subsampling rate $S = 2$, hence length of subsampled activity is truncated: $500 \rightarrow 500/2 = 250$. (C) $S = 5$, (D) $S = 10$.

4.7.1 Subsampling & smearing

A simple approximation of the affects of fMRI as supported by (Gomez et al., 2024; Boynton et al., 1996) is to apply a rolling window average to the activity, $\mathbf{x}(t)$ of size S , then sample at a rate of S :

$$\tilde{\mathbf{x}}_S(t) = \frac{1}{S} \sum_{i=0}^S \mathbf{x}(S(t+1) - i) \quad (4.7)$$

$$\tilde{\mathbf{X}}_S = \{\tilde{\mathbf{x}}_S(0), \dots, \tilde{\mathbf{x}}_S(\lfloor M/S \rfloor)\} \quad (4.8)$$

This process will simply be referred to as *Subsampling*.

Figure 4.7 demonstrates the effect of subsampling on the original activity (A). Subsampling has the affect of a low-pass filter, removing high frequency components present in the signal, see (A) $\rightarrow$ (D). The process also reduces the magnitude of the activity, as the averaging window reverts the signal to the mean.

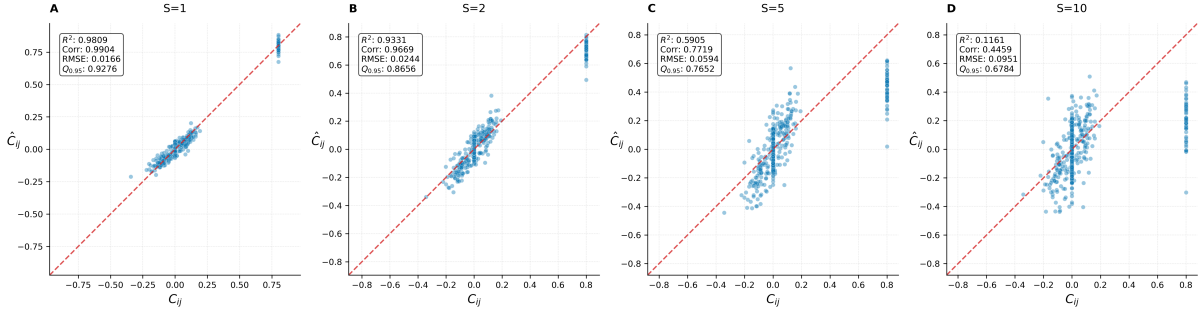


FIGURE 4.8. **Reconstructions of subsampled activity.**

Network parameters: $N = 50, k = 4, p = 0.05, \alpha = 0.8, M = 10^5$. (A) Without subsampling: $Q_{0.95} = 0.93$. (B) Subsampling rate, $S = 2$: $Q_{0.95} = 0.86$. (C) Subsampling rate, $S = 5$: $Q_{0.95} = 0.77$. (D) Subsampling rate, $S = 10$: $Q_{0.95} = 0.68$

4.7.2 Reconstruction

To test the ability for LLS to reconstruct the underlying network on subsampled activity, the original activity $\mathbf{X}$ is of an increased length: $S \times M$, which then undergoes subsampling to produce $\tilde{\mathbf{X}}$ of the intended length M . The original time series extended to separate the effects of subsampling from an insufficient observation length. Figure 4.8 describes the general effect of subsampling the activity on the reconstruction of the network. Overall when subsampling at a rate of 10 results in a drop in quality of the of reconstructions $Q_{0.95} = 0.93 \rightarrow 0.68$.

Closer inspection reveals the reconstruction overestimates the external connections (the group centred $\sim C_{ij} = 0$) and underestimates the intrinsic connections (the group centred $\sim C_{ij} = 0.8$).

This result can be reasoned: consider that the true underlying propagator, $\mathbf{C}$ acts approximately as $\mathbf{x}(t)\mathbf{C}^S = \mathbf{x}(t+S)$ (even ignoring the substantial effect of process noise being incorporated at each time step), so reconstruction of subsampled data, includes several steps between each observations. Therefore applying (4.6) to subsampled activity in fact reconstructs $\hat{\mathbf{C}}^S$. Raising the network to the power S has the effect of raising the eigenvalues of the modes of propagation: $\hat{\mathbf{C}}^S = V\Lambda^S V^{-1}$, where Λ is a diagonal matrix of the eigenvalues: λ_i^S . For $0 < |\lambda_i| < 1$ (which we have due to enforcing the stability condition), this results in: $|\lambda_i^S| < |\lambda_i|$, shrinks the modes and the magnitude of the connections. This explains how sub-sampling can reduce the apparent differences between inferred connectivity strength on connected and non-connected pairs.

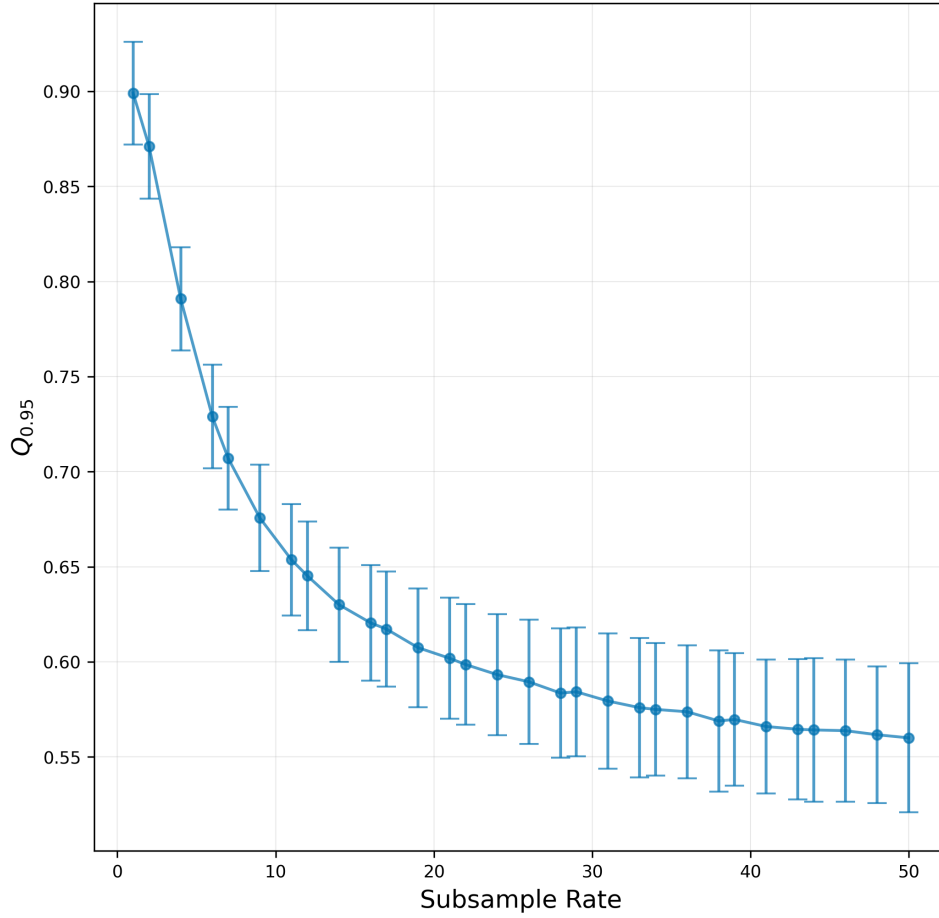


FIGURE 4.9. **Quality of Reconstruction for subsampled activity.** Network parameters: $(N = 300, k = 4, \alpha = 0.5, p = 0.05)$. Each network is reconstructed from $M = 10^3$ samples post subsampling.

4.7.2.1 Quality of subsampling

Figure 4.9 describes the affect of subsampling on the quality of reconstruction.

Figure 4.9 quantifies the decline in reconstruction quality ($Q_{0.95}$) with increasing subsampling rate. A sharp reduction in quality is observed at a subsampling factor of 10, after which the effect gradually tapers and stabilises around $Q_{0.95} \approx 0.57$.

This degradation is likely driven by the loss of high-frequency or short-timescale dynamics that capture rapid interactions between nodes. As subsampling increases, temporal smoothing and reduced sample density obscure these transient dependencies, effectively diminishing the observable information required for accurate inference.

The introduction of subsampling does not result in the complete inability to reproduce the underlying connectivity. This result suggests that at even higher subsampling rates, there is opportunity to recover a network at a $Q_{0.95} \approx 0.5$. A promising results for the applicability of LLS to fMRI data.

These results raise an important question: to what extent does the impact of subsampling depend on the characteristic frequencies and intrinsic timescales of interactions?

4.8 Fourier Transform Analysis

To investigate this dependency systematically, the frequency composition of the network activity was examined. By analysing the spectral characteristics of each node's time series, it becomes possible to identify which networks characteristics may be robust to subsampling.

A Fourier transform can obtain a spectrograph of the dominant network frequencies. The general method to do so is displayed in equation 4.9, though note actual application utilises the similar Discrete Fast Fourier Transform. The power spectral density (equation 4.10) then describes the distribution of power into the frequency components.

$$\mathbf{x}(f) = \mathcal{F}\{\mathbf{x}(t)\} = \int_{-\infty}^{\infty} \mathbf{x}(t) e^{-i2\pi ft} dt \quad (4.9)$$

$$P[\mathbf{x}(f)] = |\mathbf{x}(f)|^2 \quad (4.10)$$

For a given power spectral density the expected frequency can calculated according to: $E(P[X]) = \int_0^{\infty} |\mathbf{x}(f)|^2 f df$, which represents the average frequency component weighted by the power contribution. An example of the power spectral density of the network activity is in Figure 4.10.

4.8.1 Subsampling Frequencies

To confirm the low-pass filter nature of the rudimentary subsampling process the power spectral density of the network activity, $\mathbf{X}$ was examined post subsampling, $\tilde{\mathbf{X}}_S$ at rates of $S = 1, 2, 10$. The results are described in Figure 4.11.

Figure 4.11A confirms the suspicion, indeed the subsampling process removes higher frequency components of activity. This is indicated by the drop in power in the higher frequency regions for $P(\hat{\mathbf{X}}_2)$, $P(\hat{\mathbf{X}}_{10})$,

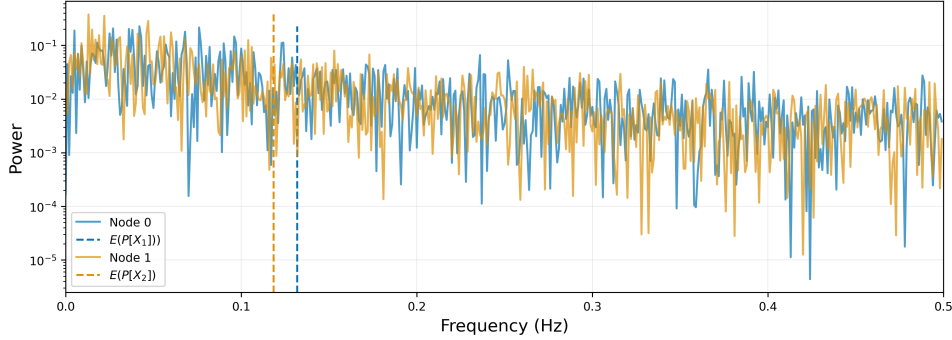


FIGURE 4.10. **Log Power spectral density of network activity.**

The power distribution of the activity of 3 nodes, calculated according to equation 4.10. At a sample of frequency $f_s = 1$, allows for resolutions of $f < f_s/2 = 0.5$ according to (4.15). Network parameters: $(N = 300, k = 4, \alpha = 0.5, p = 0.05)$, $M = 10^4$ sec. Vertical lines represent the expected frequency of the power spectrum: $E(P[\mathbf{X}])$

whilst the lower frequencies $\approx 0.01\text{Hz}$ remain unchanged. Figure 4.11B demonstrates that as the sub-sampling rate increases the expected frequency decays asymptotically.

The results from Figure 4.11 suggest that networks which mainly generate high frequency activity would become unrecoverable from heavily subsampled and filtered observations. How about networks which exhibit low frequency activity such as the observations expected of fMRI, are these networks spared from such destruction?

4.9 Intrinsic Connectivity

It is theorised that the intrinsic connectivity influences the frequencies of the generated activity, that a network with high intrinsic connections will generate low frequency activity. Investigation into the influence of the intrinsic connectivity will allow exploration of how: infra-slow activity affects reconstruction under the subsampling/fMRI process.

4.9.1 Expected Frequency

The intrinsic connectivity, α directly relates to the rate at which the activation of a node decays at each discrete time step, given $x_i(t+1) = \alpha x_i(t) + \dots$. It would be expected for networks with high α that both: the effect of extrinsic connections would be reduced due to the spectral radius constraint

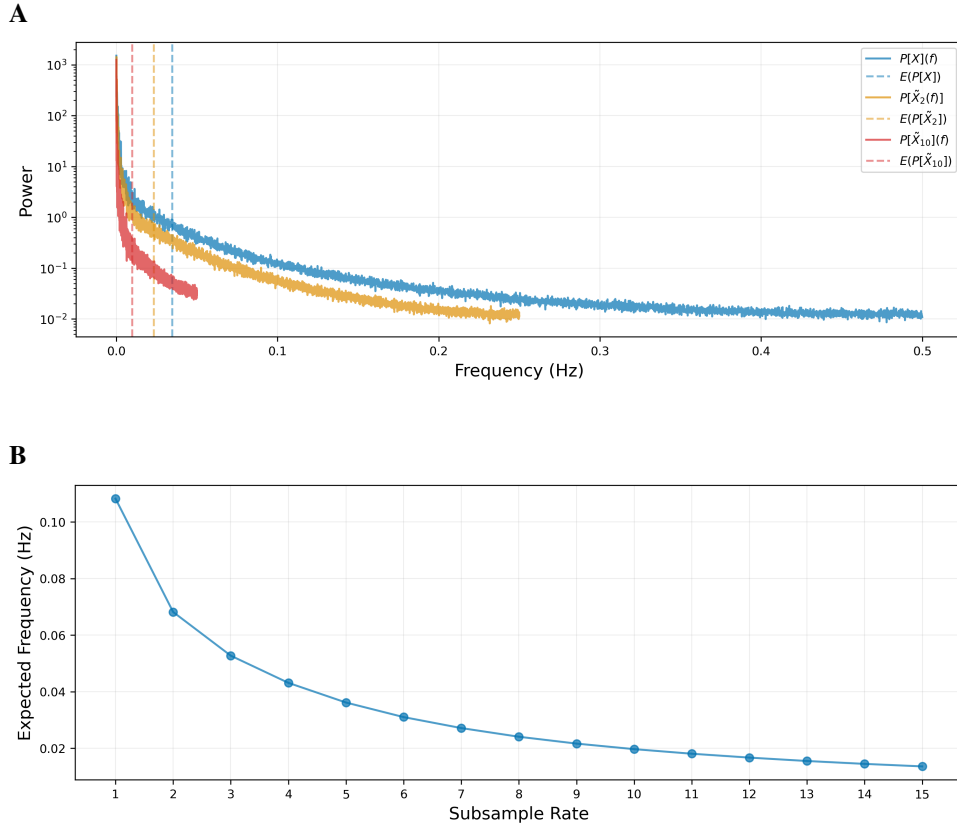


FIGURE 4.11. Power Spectral Density & Expected frequencies of Subsampled activities.

Network parameters: $(N = 300, k = 4, \alpha = 0.5, p = 0.05), M = 10^4$. (A) Average Power Spectral Density of $\tilde{X}_S$ for $S = 1$ (Original), 2, 10. Expected frequency plotted as dashed lines at: $E(P[\tilde{\mathbf{X}}]) = 0.11, E(P[\tilde{\mathbf{X}}_2]) = 0.068, E(P[\tilde{\mathbf{X}}_{10}]) = 0.019$. (B) Expected frequency of subsampled activity.

($\rho(\lambda) < 1$) and the activity of nodes would persist for longer duration. Figure 4.12 describes this relationship.

Figure 4.12 describes the theorised relationship, with increasing intrinsic connectivity the mean frequency decreases. Meaning that the wave length of generated signals are longer and remain measurable over greater sampling windows.

Combining the results from Figure 4.11B & 4.12 we may theorise that at higher subsample rates: $S > 15$ the $E(P[\tilde{\mathbf{X}}]) \approx 0.01$ which coincides with networks with an $\alpha > 0.9$. This suggests that most powerful components of activity generated from high intrinsic networks align with the observable frequencies seen at high subsampling rates.

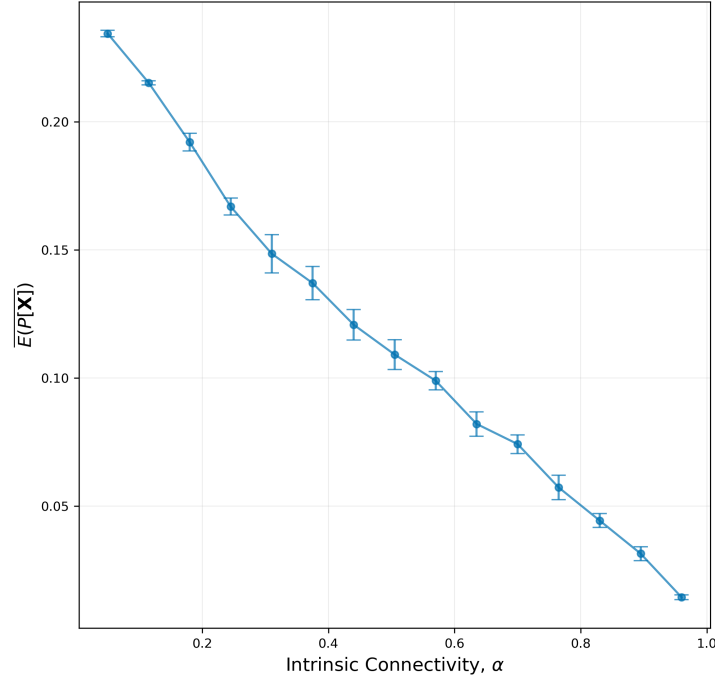


FIGURE 4.12. **Mean Expected frequency for intrinsic connectivities.**

Network Parameters: ($N = 300, k = 4, p = 0.05, \sigma = 0.3$), $M = 10^3$

The next section explores whether this alignment provides real inference performance improvements on infra slow networks ($< 0.1\text{Hz}$) under subsampling/filtering.

4.10 Infra-Slow Networks

Building on the observation that LLS inference remains partially effective under subsampling, this section investigates whether robustness improves for infra-slow networks, those whose dominant frequencies fall below 0.1Hz , characteristic of resting-state fMRI dynamics. By systematically manipulating subsampling rates, intrinsic connectivity, and frequency content, we assess how the temporal scale of network dynamics influences reconstruction accuracy and whether low-frequency activity can still yield faithful connectivity estimates after subsampling and filtering.

4.10.1 Subsampling Quality

Figure 4.13 explores the correlation between the reconstruction and underlying connectivity for a range of intrinsic connectivities, α and subsampling rates, S . As predicted, Figure 4.13B shows that for

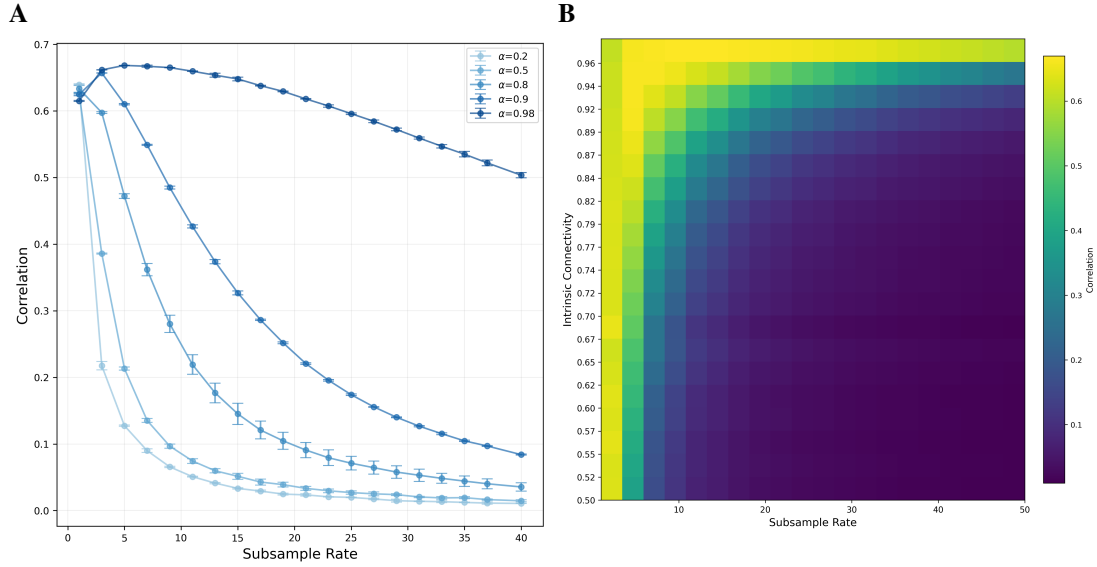


FIGURE 4.13. **Reconstruction correlation against subsampling.**

Network Parameters: $(N = 300, k = 4, p = 0.05, \sigma = 0.3)$, $M = 10^4$. Correlation was the chosen performance metric as it is agnostic to the scale changes of the extrinsic connections for differing intrinsic connectivities.

networks with $\alpha < 0.9$, reconstruction correlation declines exponentially with increasing subsampling rate, yielding poor performance beyond $S > 5$. In contrast, a sharp transition occurs for $\alpha > 0.9$, where the correlation decreases approximately linearly and remains relatively high ($r > 0.5$) even under substantial subsampling.

4.10.2 Low-pass Filtering

To test the robustness of the reconstruction technique to the removal of higher frequencies of activity from the subsampling process a low-band pass filter is applied to the Fourier transformed frequency components before performing an inverse-transform. The transformed activity can then be used to reconstruct the underlying connectivity to observe effects on quality.

Considering a network with $\alpha \approx 0.9$, where the expected frequency, $E(P[\mathbf{X}]) \approx 0.05$ (Figure 4.12), is relatively low, it is hypothesised that only the low-frequency components are required for a faithful reconstruction.

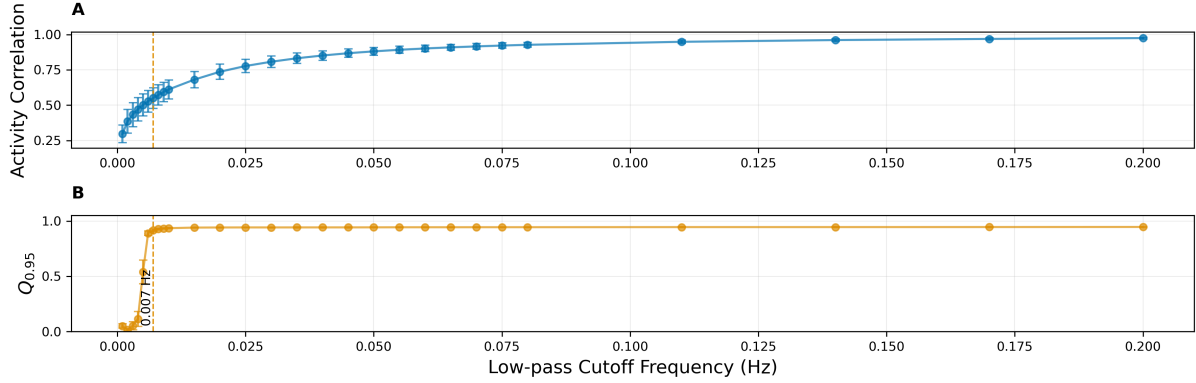


FIGURE 4.14. **Low-pass Cutoff Activity Correlation & Reconstruction Quality.**

A Low-pass cutoff filter retains frequencies from $[0, \text{cutoff frequency}]$. Network Parameters: $(N = 300, k = 4, p = 0.05, \alpha = 0.9, \sigma = 0.3), M = 10^4$. (A) Transformed activity correlation with original activity. (B) Quality of reconstruction of network inferred from transformed activity. Dotted line indicates $Q_{0.95} > 0.95$, at 0.07Hz

Figure 4.14 demonstrates that an accurate reconstruction $Q_{0.95} = 0.95$ is capable with just the low frequencies of activity $< 0.07\text{Hz}$ and ample samples, $M \gg N$. This is while the transformed activity remains only modestly correlated, $r \approx 0.5$. This suggests that a network which generates low frequencies (e.g. $E(P[\mathbf{X}]) \approx 0.05\text{Hz}$) can be reconstructed from low-pass filtered data as long as the characteristic frequencies are included.

4.10.3 Reconstructed Infra-Slow Networks

Whilst it may be possible to infer the connections from low-pass filtered activity, does the reconstructed network reflect the dynamics of the original network? To test this: the original activity, $\mathbf{X}$ of a high intrinsic network ($\alpha = 0.8$), $\mathbf{C}$ was low-pass filtered to remove frequencies $> 0.1\text{Hz}$. Both the filtered and original activity was used to reconstruct networks $\tilde{\mathbf{C}}, \hat{\mathbf{C}}$, respectively. Further activity was generated from the both reconstructed networks, resulting in: $\tilde{\mathbf{X}}, \hat{\mathbf{X}}$, respectively. This is all to analyse the effect of filtering on the resultant power spectrum of the original, reconstructed and filtered+reconstructed activities was plotted. The results are plotted in Figure 4.15.

Figure 4.15 shows that when high-frequency components are removed from the activity prior to reconstruction, the resulting network generates activity with correspondingly reduced frequencies. Because the original activity and the activity reconstructed from unfiltered data share nearly identical power spectra and expected frequencies, these changes can be attributed to the filtering process itself. This indicates that, although a network can be reconstructed from low-pass filtered activity, its inferred dynamics are

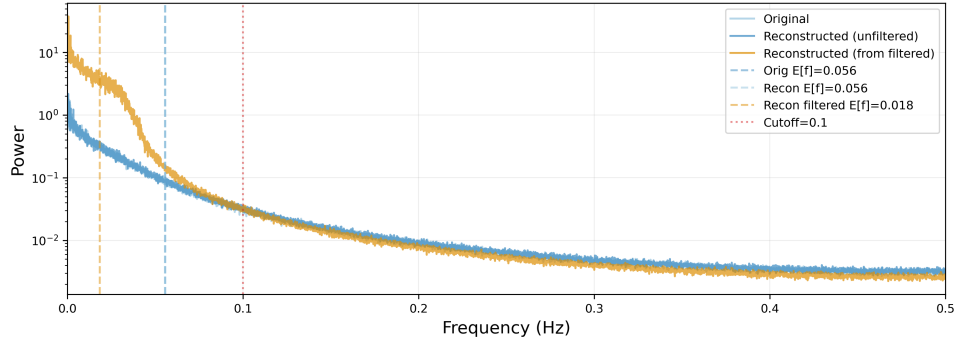


FIGURE 4.15. Average power spectrum of reconstructed networks (original & low-pass filtered).

Original activity (Light blue), Reconstructed (unfiltered) (Blue) is the activity generated from a reconstructed network, Reconstructed (from filtered) (Orange) is the activity generated from a network which as reconstructed from low-pass filtered (cutoff= 0.1Hz) activity. Network Parameters: ($N = 300, k = 4, p = 0.05, \alpha = 0.8$), $M = 10^4$. Low-pass filter cutoff is 0.1 Hz. The Original activity and Reconstructed (unfiltered) overlap entirely.

necessarily shaped by the frequency content retained during preprocessing, and the resulting connections must reflect this constraint.

4.10.4 Dependent Sampling

Previous experiments throughout this section considered reconstruction of infra-slow networks with ample samples $M = 10^4$, though experimental fMRI data has a typical length in the range of 10^3 . It's important the investigate the quality of reconstruction at this verge of $M \sim N$ with higher intrinsic connectivities.

Figure 4.16 depicts $Q_{0.95}$ alongside the activity correlation mean lifetime for increasing intrinsic connectivity with a sample length of $M = 10^3$.

The activity correlation mean lifetime describes the number of steps required for the average activity correlation between each region pair to fall to $\approx 37\%$ of their original value, assuming an exponential decay. The activity correlation mean lifetime, τ is calculated according to:

$$\text{First } \tau \text{ for which } \overline{\Omega(\tau)} < \frac{e}{\overline{\Omega(0)}} \quad (4.11)$$

Where the covariance, $\Omega(\tau)$ is calculated according to equation 4.5.

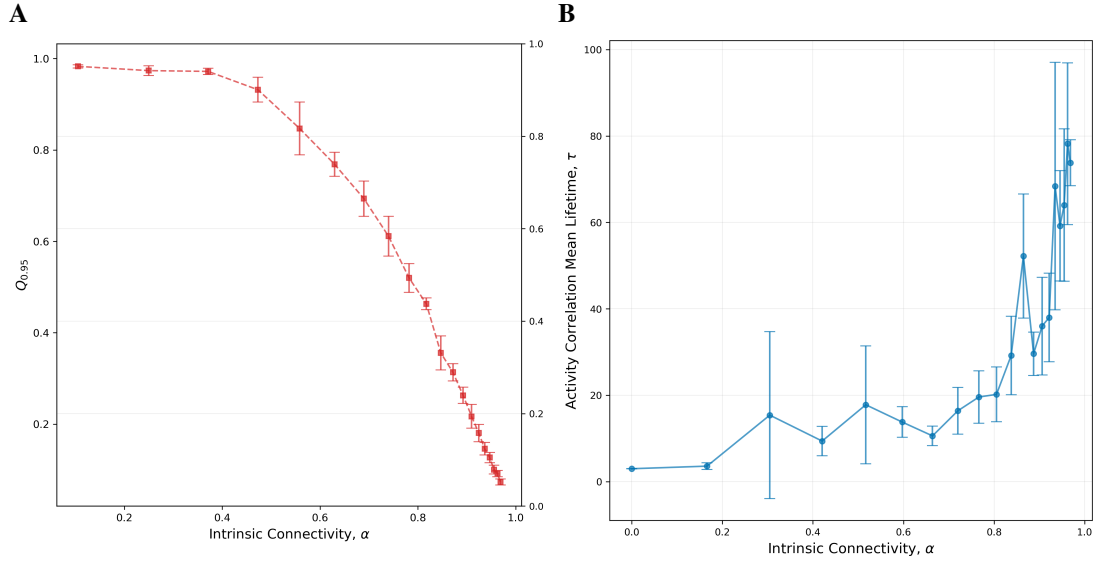


FIGURE 4.16. **Effect of intrinsic connectivity on the average correlation length (mean lifetime) and quality of reconstruction.**

Network parameters: $(N = 300, k = 4, p = 0.05, \sigma = 0.3), M = 10^3$. (A) Quality of reconstruction with increasing intrinsic connectivity, α . (B) Activity correlation mean lifetime, τ calculated according equation 4.11.

Figure 4.16 shows two results: (A) whilst the number of samples is fixed, $M = 10^3$ the quality of reconstruction will in turn drop dramatically from $Q_{0.95} \approx 0.95 \rightarrow 0.05$ as the intrinsic connectivity increases. (B) the number of steps in which nodes remain correlated increases exponentially as $\alpha \rightarrow 1$.

The results from Figure 4.16A appear to contradict the condition that when $M > N$ the quality of the reconstruction is $Q_{0.95} > 0.95$ (Figure 4.6A). This results can be reasoned by returning to the theory behind the linear least squares regression, the condition for an over-determined system of equations not only requires more known than unknowns variables but also that each known variable is independent. For high intrinsic networks the activity propagates slowly and correlation between nodes last for > 50 steps (Figure 4.16B), each successive observation is highly correlated and not necessarily independent. This reduces the exploration of the possible interactions between nodes within the recording period effectively shortening the number of samples. For case where M is of the same order as N , this results in a reduction of the quality of the reconstruction.

4.11 Connectivity Aggregation

Figure 4.16 indicates that in the infra-slow regime that reconstruction of single subject may not be possible for the length of recording provided by the HCP dataset ($M = 10^3$). Therefore it is necessary to consider how to aggregate results to "Boost" the EC and form a group representative EC.

4.11.1 Boosting Options

In order to retrieve a more complete effective connectivity, the individual subject results must be aggregated to the group level or 'boosted' to provide sufficient information. There are two mindsets of retrieving a group-level estimate that whilst they appear similar though, influence the results and interpretation:

- (1) Boost Connectivity (BC): Average the reconstructions of individual subjects.

$$\mathbf{C}_{BC} = \frac{1}{K} \sum_i^K \hat{\mathbf{C}}_i \quad (4.12)$$

- (2) Boost Activity (BA): concatenate the activity of all subjects and reconstruct a single connectivity

$$\mathbf{X}_{BA} = [\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_K]^T \rightarrow \mathbf{\Omega}_1 \mathbf{\Omega}_0^{-1} = \mathbf{C}_{BA} \quad (4.13)$$

Where there are K single subject, $\hat{\mathbf{C}}_i$ is the single-subject EC of subject i , calculated according to equation 4.6 and $\mathbf{X}_i$ is the single-subject activity before calculating the according correlation matrix.

Another option would be to average each correlation matrices: $\mathbf{\Omega}_0$ & $\mathbf{\Omega}_1$ (equation (3.11)) from each subject before reconstructing a single connectivity. Although the operations involved in equation(4.5) and the averaging are linear and therefore by the associativity property the result is the same as that of the Boost Activity.

4.11.2 Results

Experimentation of the capabilities of boosting connectivity & activity was conducted by generating HCP similar networks ($N = 300$, $M = 10^3$) with high intrinsic connectives ($\alpha = 0.9$), which display the characteristic infra-slow propagation of activity (< 0.1 Hz). Then attempting to reconstruct connectivity of a single-subject and group averages. The results (Figure 4.17) show that while a single

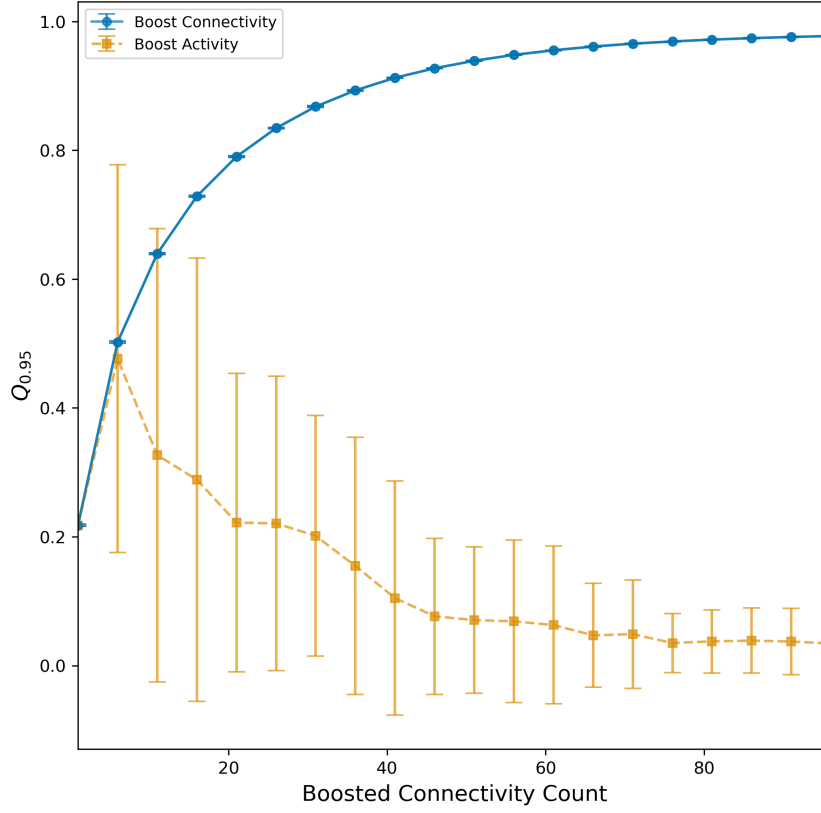


FIGURE 4.17. **Comparison of boosting activity and connectivity for independently sampled connectivities.**

Individual network's $\tilde{\mathbf{C}}$, sampled from: $\tilde{C}_{ij} = C_{ij} + \mathcal{N}(0, \overline{|\mathbf{C}|})$. Network parameters: ($N = 300, k = 4, p = 0.05, \alpha = 0.9$), $M = 10^3$.

reconstruction is poor $Q_{0.95} \approx 0.2$, the quality improves up to $Q_{0.95} \approx 0.97$, when computing the average of the 100 networks (Boost Connectivity) each generated with independent random noise applied to the 'true' population average.

The results also demonstrate that the process of Boosting Activity in fact reduces the quality of the reconstruction, $Q_{0.95} : 0.2 \rightarrow 0.05$. LLS regression requires that time series is normally and independently distributed. In this case each of the additional boosted segment of activity is generated from differing networks, hence the noise introduced is not independent sample to sample.

Although the Boost Activity process was capable of approximating the diagonal values, $\overline{\tilde{\mathbf{C}}_{BC,ii}} = 0.94$, which is the most consistent feature across all the samples.

4.12 Summary

This chapter presented many statistical observations regarding the reconstruction quality of LLS and the influencing factors. A brief summary of them is as follows.

Practical regime for linear least squares. First verified the precondition for successful recovery in this framework: an over-determined system. Empirically, quality collapsed whenever $M \leq N$, while $M \gtrsim 3N$ produced high quality reconstructions (e.g., $Q_{0.95} \rightarrow 0.95$; Figure 4.6).

fMRI as a low-pass observation model. To approximate the temporal smoothing imposed by the haemodynamic response and sampling (i.e., fMRI acting as a low-pass/rolling-average measurement), subsampling was applied and windowed averaging (equation 4.8). As expected, this procedure attenuated high-frequency components of the simulated activity (Figure 4.11), lowering the expected frequency of the spectrum and reducing reconstruction quality (Figure 4.9). Consequently, networks whose interactions generate high-frequency fluctuations are systematically underestimated or lost, whereas slowly evolving interactions remain recoverable at coarser sampling.

Role of intrinsic connectivity. Varying the intrinsic term α revealed a tight link between timescale and recoverability. Larger α slowed the dynamics (lower expected frequency; Figure 4.12), making those networks more robust to subsampling. Accordingly, reconstruction accuracy decayed steeply with subsampling for $\alpha < 0.9$, but degraded only gradually for $\alpha \geq 0.9$ (Figure 4.13). Thus, low-frequency networks are preferentially inferable from fMRI-like observations, whereas fast networks require finer sampling to be recovered.

Fourier analyses confirmed that subsampling removes energy at higher frequencies while leaving infra-slow content largely intact (Figure 4.11). Consistent with this, accurate recovery was possible from low-pass-filtered activity provided the retained band encompassed the network's characteristic frequencies (Figure 4.14). Ultimately the inferred $\hat{\mathbf{C}}$ represents a slow-propagation surrogate of the original system pre filtering.

Dependence reduces the effective sample size. At high α , activity exhibits long correlation lifetimes (Figure 4.16B), violating the independence assumption of ordinary least squares. Although M may exceed N nominally, consecutive samples become strongly dependent, shrinking the effective number of equations. This explains the apparent paradox that quality can fall sharply with increasing α at

fixed M (Figure 4.16A), despite satisfying $M > N$: the system becomes under-informative in practice because slow dynamics limit exploration of the state space within the recording window.

Implications for empirical fMRI and boosting. Given typical resting-state acquisitions ($M \sim 10^3$) and infra-slow ($\alpha \sim 0.9$) BOLD dynamics, single-subject recovery of a full directed, weighted $\mathbf{C}$ is unreliable. We therefore evaluated aggregation strategies. Averaging connectivities across subjects (Boost Connectivity; BC) improved estimates monotonically with cohort size, approaching ground truth as independent noise canceled out (Fig. 4.17). In contrast, concatenating activity across heterogeneous subjects (Boost Activity; BA) degraded performance, consistent with LLS assumptions: mixing non-identically distributed segments inflates structured noise and further violates independence.

4.13 Discussion

4.13.1 Observation Length

Shandilya and Timme (2011) further examined how observation length (sample size) and sampling rate influence the quality of network reconstruction. Their study applied a similar linear least-squares inference approach, but to data sampled from an underlying continuous dynamical process. Notably, they reported that the required observation time scales sub-linearly with network size and remains robust even in the presence of substantial noise (an effect not explored in the present investigation). These findings indicate that the number of required observations does not increase prohibitively with network size, suggesting that the method remains practical for inferring connectivity in larger systems, provided that sufficient samples are available.

4.13.2 Limits of sampling

In respect to Fourier transform (equations 4.9) the Nyquist-Shannon sampling theorem, denotes that the maximum frequency, f_{max} that can be perfectly recovered loss (without loss to aliasing) from a signal sampled at the frequency f_s is:

$$f_{max} < f_s/2 \quad (4.14)$$

$$\implies \lambda_{min} = 2/f_s \quad (4.15)$$

This enforces a fundamental limit on the highest frequency components that can be observed when sampling. This restriction is particularly harmful to analysis once considering subsampling reduces the sample rate and therefore making higher frequencies unobservable. See Figure 4.11A, as the subsample rate, S increases the $f_{max} = 1/S$ decreases. This asymptotically restricts the underlying high frequencies of activity preventing, removing them from the inference process. Generally fMRI sample periods are $TRs \sim 0.73\text{sec}$, $\implies f_s \sim 1.4\text{Hz}$, which restricts the maximum observable frequency, $f_{max} < 0.68\text{Hz}$.

4.13.3 Infra-slow Network

The results from Figure 4.15 imply that a reconstructed network generates activity that necessarily reflects the processes that transformed the source activity. This may appear obvious but the implication confines the biological model that can be produced from infra-slow fMRI activity. Assuming a reliable and faithful reconstruction of the brain can be inferred from fMRI data, this brain connectome is necessarily one that operates at a low-frequency.

There are several brain processes that operate at low-frequencies ($< 0.1\text{Hz}$) which may be captured by fMRI. The resting-state network fluctuations incorporate the major functional modules such as the default mode network, salience and dorsal attention. Watson (2018) summarises the cognitive contributions of infra-slow oscillations and relate it heavily to role of information processing, priming the brain to attune to certain input.

The higher end of frequencies present in the brain are often grouped into bands such as alpha ($\sim 10\text{Hz}$), beta ($\sim 15 - 30\text{Hz}$) and gamma ($> 30\text{Hz}$). For example the Beta Band is prominent in the sensorimotor cortex and the frontal regions and is associated with motor control and precedes motor movement initiation (Baker, 2007). These frequencies are well above the temporal resolution of fMRI and hence are undetectable and are commonly studied with electroencephalography (EEG). Hence the proceeding study will be unable to observe and by extension to model the interactions generating these higher bands.

Although there is evidence to suggest that the infra-slow oscillations interact and modulate the power of faster oscillations, promoting and attenuating in different phases of the cycle (Watson, 2018; Ko et al., 2011). The most consistent demonstration is the modulation of the delta (0.5 - 4 Hz) and sigma (10 - 20Hz) bands by the default mode network which are inturn directors of nonREM sleep. Though there is yet a clear process on how these frequencies interact.

Hence while higher frequencies may not be observable there is still a wealth of functional information provided by the infra-slow fMRI activity that describe larger systemic transitions in the brain.

4.13.4 Dependent Sampling

The slow time evolution of activity produced by networks modeled with high intrinsic connectivity leads successive observations being highly correlated (Figure 4.16B) and statistical dependent. Figure 4.16A demonstrates that this effectively shortens number of samples, prohibiting highly quality reconstructions for limited number of for a limited number of observations $M = 10^3$. Shandilya and Timme (2011) ascribe higher reconstruction qualities at lower sampling frequencies for networks which display irregular and erratic behaviour, citing that "lower degree of correlation among observations at different times that is required for accurate reconstruction numerics, e.g. for irregular dynamics, sample points more distant in time provide more relevant information increase about the system as they are less correlated than closer-by points."

4.13.5 Boosting Activity vs Boosting Connectivity

Under the assumption that inter-subject variability is independently and normally distributed, boosting connectivity (averaging over all subject connections) produces a representative group average that is robust to noise generated from incomplete reconstructions (Figure 4.17). Boosting the activity produces a single time series with $M \gg N$, in the aim of improving the reconstruction. Though each concatenated time series display dependent variability which violates the underlying assumptions of OLS, resulting in a decreased performance for an increased boost count. Although Boost Activity will continue to be considered in upcoming investigation, as the as it provides a great statistical comparison to the BC alternative.

Brain Activity: HCP Dataset

Estimating effective connectivity (EC) at the whole-brain level enables an understanding of how spontaneous activity propagates across large-scale functional systems such as the default-mode, control, and attention networks. This provides insights beyond the undirected associations captured by functional connectivity (FC). By introducing directionality, EC reveals the hierarchical organisation of brain dynamics and identifies potential sources and sinks of neural influence across the known functional modules.

The synthetic results presented in Chapter 4 demonstrated that linear least squares (LLS) regression can accurately reconstruct directed network structures when applied to aggregated data. Group-level averaging of inferred connectivities (Boost Connectivity; BC) yielded stable and realistic estimates. Building on these findings, this chapter applies LLS inference to group-averaged rs-fMRI data to estimate a whole-brain effective connectivity.

5.1 Method Outline

In order to generate a robust analysis and interpretation of the emergent directed EC components across the whole brain, the EC inference is supplemented with the following steps:

- (1) **Whole-brain EC inference:** Apply the validated LLS technique to real rs-fMRI data to reconstruct the effective connectivity of the human brain.
- (2) **Signal enhancement and dimensionality reduction:** Improve the signal-to-noise ratio of reconstructed connectivity matrices and extract the most significant components for analysis.
- (3) **Model evaluation:** Quantify the fit and predictive performance of the inferred model relative to observed data.

- (4) **Network interpretation:** Analyse the resulting connectivity structure to identify major organisational features of the brain and discuss their functional and theoretical implications.
- (5) **Delay Network Modelling:** Extend the current network model to include time-lag activity and discuss the implications of the sense derivative propagator

This is all to answer the question:

When applied to resting-state fMRI recordings, what effective connectivity structure emerges at the whole-brain level, and what are its potential biological interpretations?

5.2 Preliminary Analysis

Preprocessing

Before inference, each subject's BOLD time series is preprocessed to remove the DC component, detrended, and low-pass filtered with a zero-lag order-3 Butterworth filter (0.01–0.08 Hz). To mitigate filter initialisation effects, 100 samples are removed from the beginning and end of each time series (Cliff, 2021). These steps reduce artefacts related to head motion, physiological noise (e.g., respiration), and differences in acquisition timing across regions (Glover, 2011).

Functional connectivity check

As a basic validation, a sample functional connectivity (FC) matrix is computed as the time-average correlation between regional time series (equation 3.10). Subject-level FC matrices are then averaged to obtain a population-representative FC. Figure 5.1 shows this average FC both in Gordon parcellation order and grouped by Yeo's functional modules (Yeo et al., 2011).

Gordon's parcellation groups the left and right hemisphere's of the brain together, to this effect Figure 5.1A displays strong intra-hemispheric correlations (second and fourth quadrants) and inter-hemispheric correlations (first and third quadrants) between homologous networks. Figure 5.1B describes Yeo's 17-network scheme revealing strong intra-module correlation within the visual, somatomotor, ventral attention and default networks for example.

The clear alignment of the calculated functional connectivity with the known functional modules indicate that brain activity within the HCP rs-fMRI contains information regarding the region-region correlation.

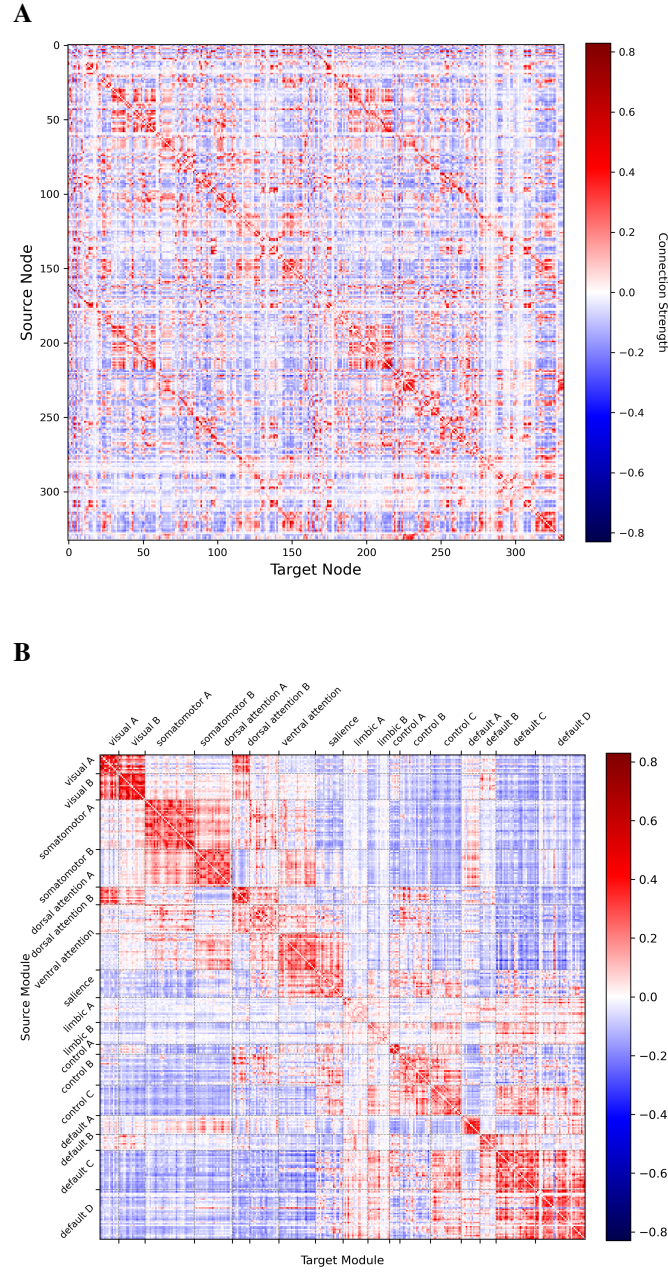


FIGURE 5.1. Brain Functional Connectivity & Yeo's Modules.
 (A) The average functional connectivity in Gordon's parcellation order. (B) The average functional connectivity grouped in Yeo's modules with corresponding functional labels.

Single-subject Reconstruction

Before completing a group representation reconstruction it worth constructing a single-subject EC to evaluate the LLS inference ability to produce a reliable network with $M = 10^3$ observations. Applying

the LLS inference (equation 4.6) to a single subject's activity $\mathbf{X}_i \in \mathbb{R}^{N \times M}$, with $N = 333$ and $M = 10^3$ results in Figure 5.2.

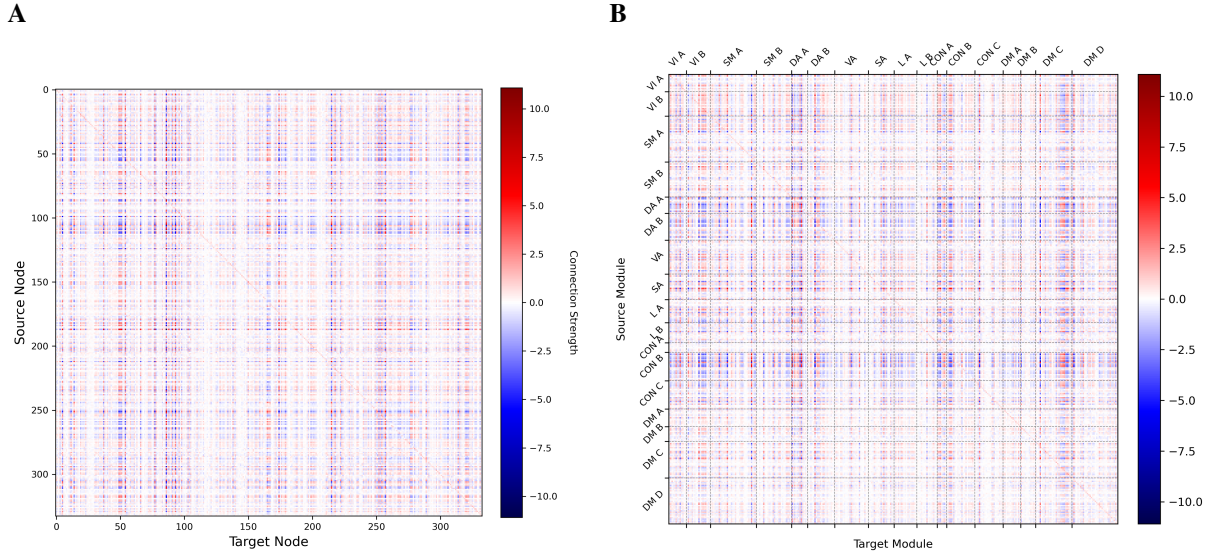


FIGURE 5.2. **Effective Connectivity of single subject.**

$N = 333$, $M = 10^3$. Mean diagonal, $\overline{C}_{ii} = 0.81$

Figure 5.2 shows that only a partial reconstruction of the effective connectivity could be achieved, even though the sampling condition $M > N$ was satisfied. The HCP resting-state fMRI dataset provides a single 15-minute scan per subject, sampled at a TR of 0.73 s, yielding approximately 1,200 time points per individual. Such a duration is insufficient to accurately infer single-subject effective connectivity using linear least squares regression. Consequently, the reconstructed networks are sparse and incomplete.

This suggests that a single resting-state fMRI time series remains insufficient for a complete network reconstruction, a result that was consistent across subjects.

5.2.0.1 Discussion

Previous research examining the scan durations required for reliable single-subject functional connectivity has demonstrated observation lengths < 10 min, once considered adequate (?), produce inconsistent and unstable connectivity estimates (Laumann et al., 2015). Birn et al. (2013) showed that extending scan durations beyond 15 minutes markedly improves the reliability of individual connectivity estimates, enabling more robust differentiation between individuals and group averages (Anderson et al.,

2011). Given that effective connectivity entails estimating a directed (asymmetric) network, effectively doubling the number of parameters relative to undirected functional connectivity, it is reasonable to expect that comparable reliability would require at least twice the recording length, on the order of ≈ 30 minutes, to achieve similar discriminatory power.

A comparable scenario was examined in the synthetic experiments (Figure 4.16), where networks with high intrinsic connectivity ($\bar{C}_{ii} \approx 0.81$) exhibited slow temporal dynamics and strongly correlated successive samples. Under these conditions, the expected reconstruction quality was reduced to $Q_{0.95} \approx 0.35$. The similarity between these synthetic and empirical results suggests that intrinsic network dynamics, rather than sample size alone, fundamentally limit single-subject reconstructions from resting-state fMRI data.

A closer examination of the power spectral density of the activity place the data squarely in the infra-slow regime (<0.1 Hz) (Figure 5.3). As a result, adjacent observations are strongly autocorrelated rather than independent. The effective number of independent samples is therefore much smaller than the nominal length M .

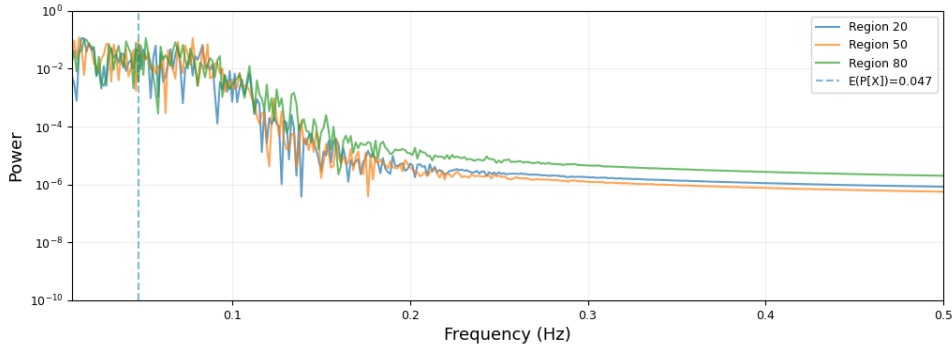


FIGURE 5.3. Power spectral density of activity.

5.3 Boosting: Group representation

As theorised in Figure 4.16A, it is not possible to produce a complete $\hat{C}$, from $M = 10^4$, $N \approx 300$, $\alpha \approx 0.81$. Hence it is necessary to aggregate data to produce a more reliable group representation. Section 4.11 included demonstrations of two boosting methods to improve reconstruction quality by determining a group-level estimate, via equations 4.12 & 4.13. This section will now look into the application of Boosting on the empirical data and the statistical interpretation of the results.

5.3.1 Interpretation Comparison

Generally when estimating how a random variable, Y depends on the X across multiple individuals, two strategies exist: Fixed effect (Boost Activity) and Random effect (Boost Connectivity) models (Dettori et al., 2022). Fixed effect models assume that each individual X share the same underlying parameters, whilst Random models assumes variance within the population.

Boosting the connectivity therefore assumes that there is meaningful variation within the population, and that each individual's estimated connectivity represents a sample from this distribution. The averaged result reflects the expected connectivity across subjects:

- It assumes that the mean network is a representative sample from the underlying population distribution.
- It expresses the most likely or common set of connections present across subjects.
- If the noise introduced through partial reconstruction is random and independent it can be expected to be reduced through averaging across the $K = 471$ samples.

In contrast, boosting the activity assumes that all individuals share the same underlying network structure and can be treated as repeated observations of a single stationary system:

- It assumes that each individual's dynamics arise from an identical connectivity structure.
- It expresses only those connections that are consistently present across the entire population, while variable connections are suppressed.
- Inter-individual diversity is treated as noise rather than a genuine signal, potentially violating the assumption temporally independent noise.

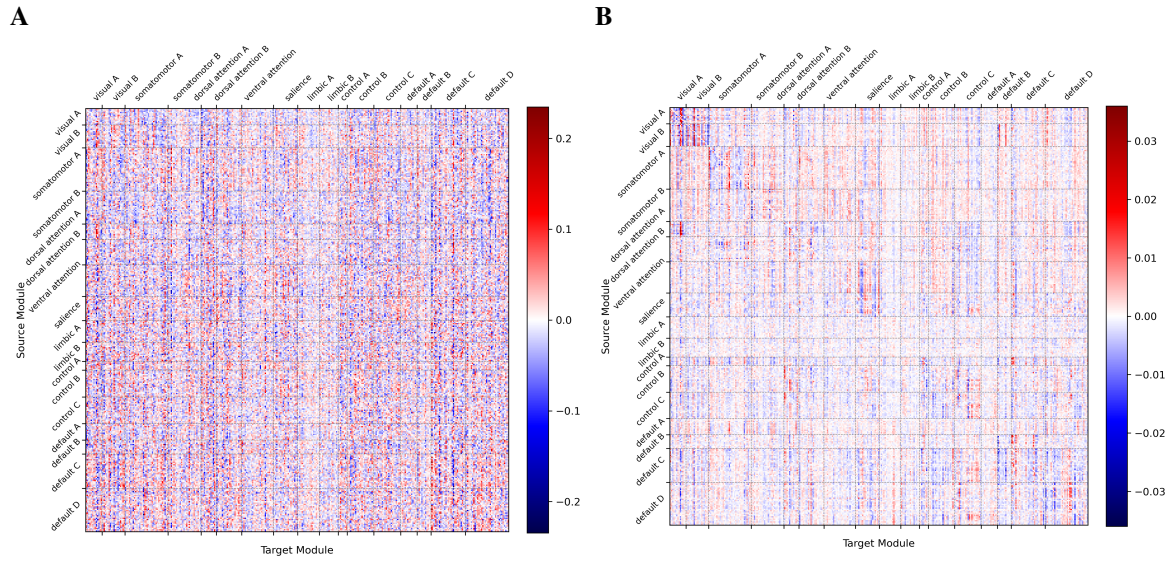
Hence, while both methods yield a group-level connectivity, they differ in interpretation. *Boost Connectivity* (BC) captures the *expected* network across a heterogeneous population, whereas *Boost Activity* (BA) estimates a homogenised population, with a network dominated by the most consistent, stationary connections.

5.3.2 Group Effective Connectivity Results

This chapter will present the results of applying both BA and BC process given by equations 4.13&4.12, as there is meaningful conclusion to be drawn from the differences between the networks.

The results of group aggregation are in figures 5.4A & 5.4B.

For the remainder of the report the diagonals of the connectivity matrix plots (intrinsic connections) will be omitted for visually clarity. Generally the intrinsic connections are 10^2 times greater than the inter-region connections, suppressing the contrast. Where appropriate the diagonal values will be stated alongside the plots.



Parameter	Boost Connectivity		Boost Activity	
	Mean	Std	Mean	Std
$\mathbf{C}$	2.8×10^{-3}	58×10^{-3}	2.9×10^{-3}	53×10^{-3}
$ \mathbf{C}_{ij} $	2.8×10^{-2}	2.3×10^{-2}	2.2×10^{-3}	2.2×10^{-3}
$\mathbf{C}_{ii}$	0.81	0.044	0.97	0.0075

FIGURE 5.4. **Group-level Effective Connectivity resulting from Boost Connectivity & Activity approaches.**

(A) Boosted Connectivity ($\mathbf{C}_{BC}$) represents the effective connectivity matrix obtained by averaging the reconstructed connectivities of individual subjects. (B) Boosted Activity ($\mathbf{C}_{BA}$) represents the effective connectivity estimated from concatenated regional activity across all subjects. The table summarises key network statistics: $\mathbf{C}$ denotes the overall mean connectivity across all pairs of regions, $|\mathbf{C}_{ij}|$ indicates the mean absolute coupling between distinct regions ($i \neq j$), and $\mathbf{C}_{ii}$ corresponds to the mean intrinsic (self) connectivity of each region.

There are several results in Figure 5.4 to unpack: Both boosting processes result in approximately the same average connectivity: $\overline{\mathbf{C}}_{BC} \approx \overline{\mathbf{C}}_{BA}$, suggesting that this is a consistent result across all subjects. Both $\mathbf{C}_{BC}$ and $\mathbf{C}_{BA}$ show strong intrinsic connections $\alpha = \mathbf{C}_{ii} > 0.8$, consistent with the low-frequency described in Figure 5.3. Inter-region couplings are larger in BC than BA by roughly an order of magnitude, mirroring the comparatively lower BC diagonals ($\overline{\mathbf{C}}_{BC,ii} < \overline{\mathbf{C}}_{BA,ii}$). This suggests

that, while intrinsic connectivity is on average ≈ 0.81 for an individual, BA has identified that a strong intrinsic connectivity is a significant and consistent predictor of the activity across all subjects. Qualitatively, BA yields a clear alignment with the functional modules (e.g., Visual A/B), while BC appears noisier with fewer visible structures. Though a common feature between them is the weak incoming connectivity to *limbic A* and *limbic B*.

While the previous results demonstrate the whole-brain EC the results warrant further interpretation and verification. The reconstructed networks should be examined for key structural features and tested for consistency with the underlying model assumptions, as explored in the following sections.

5.4 Results Validation

Having established a whole-brain effective connectivity from resting-state fMRI data, it is essential to verify that the inferred structure reflects genuine neural interactions rather than artefacts of noise or model assumptions. The following validation analyses test the robustness and statistical significance of the reconstructed networks.

5.4.1 Surrogate Time Series

Surrogate testing is a common check in neuroscience for testing whether the observed activity exhibits non-trivial structure (e.g. casual relationships) beyond what could have been expected by a null model. To check whether a particular connection arises from statistically significant interactions/correlations between region pairs, surrogate time series are generated where activity of individual regions are separately and randomly temporally shifted, $\mathbf{x}(t)_i \rightarrow \mathbf{x}_i(t + \gamma)$, $\gamma \sim \mathcal{U}(0, M)$, in a circular fashion. This removes any true interactions between regions from being captured in the null model constructed from the surrogate, whilst conserving intrinsic interactions; any other interactions which are captured in the null model will be merely due to statistical fluctuations.

These surrogate times series are used to generate the ECs via BA & BC. The null model connectivity is produced by averaging a sample of 30 ECs. The distribution of non-intrinsic connections in the group-level null models tell us the range of connectivities which are due to statistical fluctuations alone. The resulting average null model connectivity and it's standard deviation for BA is displayed Figure 5.5.

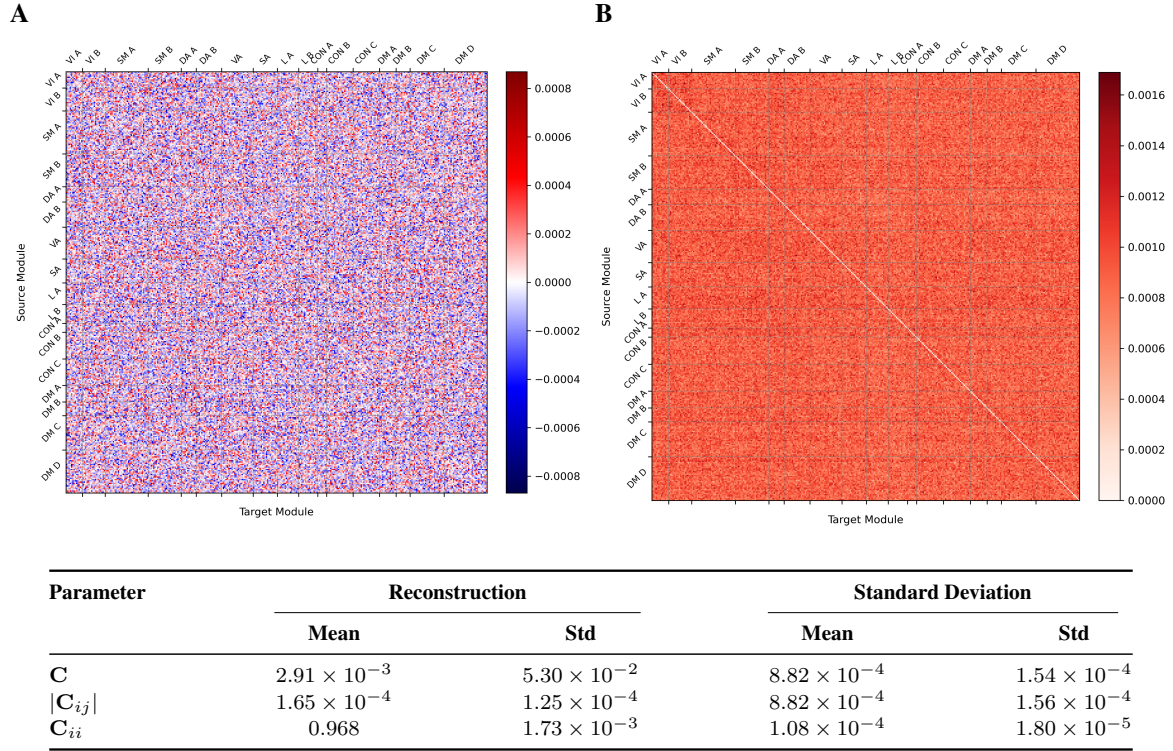


FIGURE 5.5. **BA average null network and standard deviation.**

(A) Average null model EC obtained by randomly time-shifting regional activity to remove temporal correlations. (B) The corresponding standard deviation matrix summarises variability across connections.

Visually the connections within Figure 5.5A appear as purely random noise. The strength of the inter-region connections is on average significantly less than the intrinsic connections, $|\overline{\mathbf{C}_{ij}}| = 1.6 \times 10^{-4} \ll |\overline{\mathbf{C}_{ii}}| = 0.97$. This implies that with the removal of temporal relations between the regions of the brain, the coupling interactions become insignificant in comparison the auto regressive self interactions. The result for the null network constructed via BC is similar.

5.4.2 Significant Connections

Statistically significant connections between brain regions can be identified as those that arise from non-trivial interactions, determined by comparing the reconstructed connectivity of BA & BC against the null model.

To reduce spurious type-I errors (false-positive) when conducting multiple testing against each of the 333×333 possible connections Bonferroni correction is applied. For a statistical significance of $p <$

0.05, the corrected $p_{\text{corrected}} < 0.05/(333^2) = 4.5 \times 10^{-7}$. Hence connections are only included if their magnitude exceeds a z score of ≈ 5 (corresponding with $p_{\text{corrected}}$) away from the average null model, indicating that their presence cannot be explained by more than chance of uncorrelated regional activity. Connections in which are not considered significant are removed from both the BC and BA connectivities. The results are described in Figure 5.6A.

After filtering for significant connections, BA EC connectivity drops from 99.65% to 13.18% (a $\sim 86\%$ reduction) the BC EC connectivity drops from 99.97% to 88.76% (an $\sim 11\%$ reduction).

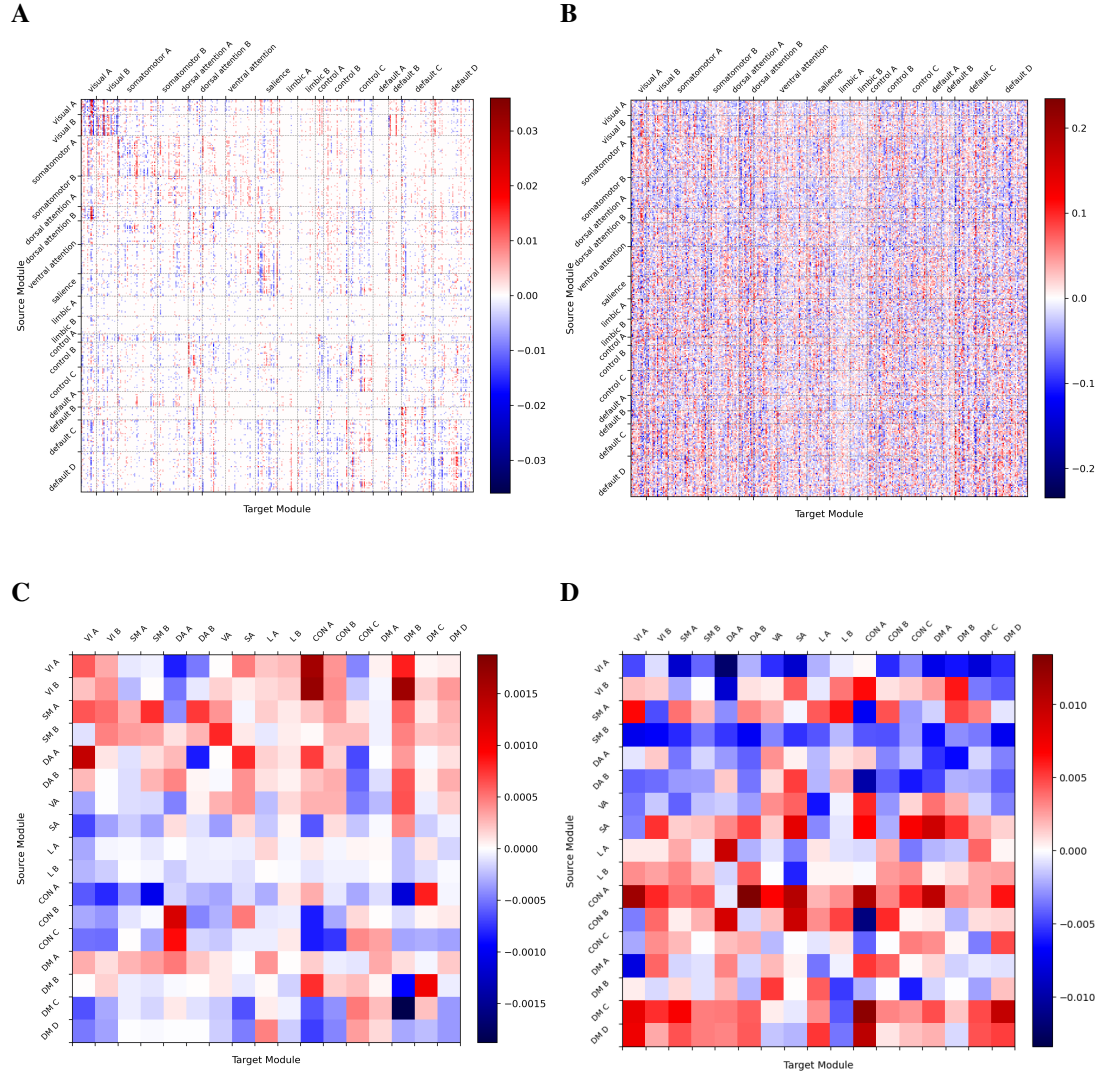
For BA, the significant reduction in the number of surviving connections indicates that majority of the original connections may be due to random fluctuations. This may also be due to inter-subject variability, effectively competing with each other during fitting the model to all subjects simultaneously, blurring any finer interaction. Though the small set of remaining edges can be considered statistically significant across all subjects.

For BA a strong majority of connections remained after filtering, demonstrating that the process efficiently identifies statistically significant group representative connections

Plotted alongside the filtered ECs is their respective functional module averages, Figure 5.6C&5.6D. This allows identification of average modular interactions. The averaging was performed within each of the module pairs, excluding the intrinsic connections, to avoid obscuring the intra-module connections.

5.4.3 Network Model: Goodness of fit

As part of the model-based inference framework, the reconstructed connectivity matrix is fitted to explain the observed activity dynamics, under the assumption that the model's generative structure reflects the underlying network organisation. To assess the validity of the model's assumptions (e.g., linearity, multivariate interactions, and stationarity) and to demonstrate the model's effectiveness relative to simpler or misspecified alternatives, the predictive capacity of the inferred connectivity can be evaluated by quantifying how much variance in the empirical brain activity it is able to explain.



Parameter	Significant BA EC Matrix		Significant BC EC Matrix	
	Mean	Std	Mean	Std
C	2.61×10^{-3}	5.19×10^{-2}	2.84×10^{-3}	57.7×10^{-3}
$ C_{ij} $	7.5×10^{-4}	2.24×10^{-3}	2.5×10^{-2}	2.6×10^{-2}
C_{ii}	0.86	0.296	0.81	0.04

FIGURE 5.6. **Significant BA Effective Connectivity.**

(A) BA Significant effective connections ($p < 0.05/333^2$, corresponding to $|z| > 5$). (B) BC Significant effective connections ($p < 0.05/333^2$, corresponding to $|z| > 5$). (C) BA Module-level summary showing the average connectivity between Yeo's functional networks. This excludes the intrinsic connectivities. (D) BC Module-level summary showing the average connectivity between Yeo's functional networks.

With the inferred connectivity, $\hat{\mathbf{C}}$ the predicted activity, $\hat{\mathbf{X}}$ can be calculated according to equation (5.1). The proportion of the variation in activity explained, R_i^2 for each brain region i by the inferred connections, $\hat{\mathbf{C}}_{ji}$ can then be quantified according to equation (5.2).

$$\hat{\mathbf{X}}(t+1) = \mathbf{X}(t)\hat{\mathbf{C}} \quad (5.1)$$

$$R_i^2 = 1 - \frac{\sum_{t=0}^{M-1} (\mathbf{x}_i(t+1) - \mathbf{x}(t)\hat{\mathbf{C}}_{:,i})^2}{\sum_{t=0}^{M-1} (\mathbf{x}_i(t+1) - \bar{\mathbf{x}}_i)^2} \quad (5.2)$$

where $\hat{\mathbf{C}}_{:,i}$ is the i th column of the inferred connectivity, relating to the incoming connections to region i and $\bar{\mathbf{x}}_i$ is the temporal mean of the recorded activity for region i . The predictive accuracy of the model is quantified by the proportion of variance in the empirical activity that can be explained by the model predictions, R^2 . Note that the process requires sufficient observations to produce a complete connectivity for stable activity prediction therefore BC could not be applied in this study as there is no method to produce a reliable single-subject EC. Hence this study only evaluates the fit of the group representative BA EC.

For comparison, two additional models were evaluated for their predictive power:

- **Autoregressive model:** predicts activity solely based on the previous time point, $\hat{\mathbf{X}}_{\text{AR}}(t+1) = \mathbf{A}\mathbf{X}(t)$, $\mathbf{x}_i(t+1) = a \cdot \mathbf{x}_i(t)$ where $\mathbf{A} \in \mathbb{R}^{N \times 1}$ contains coefficients estimated independently of the reconstructed connectivity.
- **Null model:** uses surrogate time-series generated by circularly shifting the original activity in time, destroying temporal correspondence while preserving autocorrelation structure.

Higher R^2 values indicate greater predictive power. Comparing R_{network}^2 against R_{AR}^2 and R_{null}^2 assesses whether the reconstructed connectivity captures meaningful network structure beyond simple autoregressive or random dynamics. The results are in Figure 5.7.

Figure 5.7A demonstrates that the network model (Red) on average is capable of explaining majority of the variation in activity, $\overline{R^2} = 0.930 \pm 0.0034$, for all regions of the brain. The AR (Blue) and null model (Grey) both demonstrate high predictive accuracy, marginally better than the network model. Figure 5.7C describes that the addition of network connections in fact reduces performance in comparison, $\Delta R_{\text{AR}}^2 = -9.5 \times 10^{-3}$, $\Delta R_{\text{null}}^2 = -8.7 \times 10^{-3}$. The high predictive ability of both the AR and null model is in part due to the comparatively significant contribution of the intrinsic connectivities of the BA EC, $\overline{\mathbf{C}_{BA,ii}} = 0.96 \gg \overline{\mathbf{C}_{BA,ij}} = 2.23 \times 10^{-3}$ (Figure 5.4B).

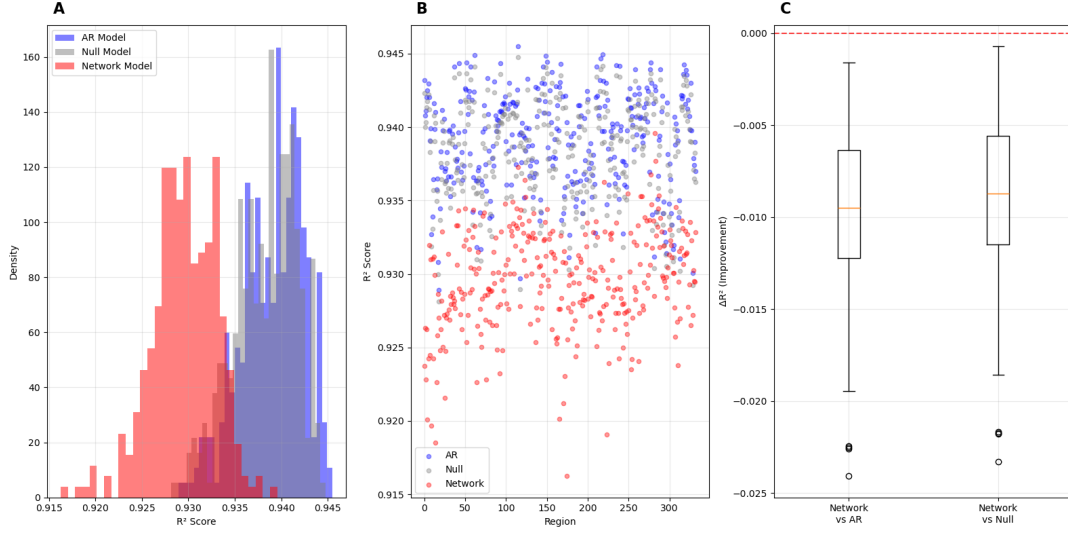


FIGURE 5.7. **Proportion of brain activity variance explained by model predictions.**

(A) Frequency density of regions R^2 scores for each model: network (Red), AR (Blue), null (Grey). The average R^2 scores; $\overline{R^2}_{\text{network}} = 0.930 \pm 0.0034$, $\overline{R^2}_{\text{AR}} = 0.939 \pm 0.0033$, $\overline{R^2}_{\text{null}} = 0.938 \pm 0.0034$. (B) Scatter of each regions R^2 score. (C) ΔR^2 improvement of the network models to alternative models: $\Delta R^2_{\text{AR}} = -9.5 \times 10^{-3} (p \ll 0.05)$, $\Delta R^2_{\text{null}} = -8.7 \times 10^{-3} (p \ll 0.05)$

5.4.3.1 Discussion

To demonstrate the advantages of the network model, it is compared to an autoregressive model (each region is a function of only itself) and a null model (underlying activity is stripped of causal relationships). The results from Figure 5.7 demonstrated that while all models explained $> 90\%$ of the variance the autoregressive and null model performed nearly 1% better. The slow propagation of activity ($E[P(\mathbf{X})] < 0.05$) leads to highly correlated subsequent activities, which allow self-coupling present in all three models to make high accurate forecasts of the horizon one step ahead.

The improved performance by the removal of the meaningful connections between regions indicates either that they 'provide confounding information' or that there is an issue within the inference technique process. As demonstrated by Figure 4.17, regressing over activity generated from multiple independently drawn connectivities (boost activity) poorly reconstructs the underlying connectivity. Therefore optimising the region-region connections to explain all the subject specific variability in activity reduced the performance in comparison to simply fitting intrinsic connections which by measure of magnitude are the most influential feature.

Due to the poor performance of the BA EC in both the model-fit and the synthetic experiment (Figure 4.17), it will be disregarded in the following interpretation of module components. of Future investigations should determine a method to quantify the model fit of the BC process, as it remains to be scrutinised.

5.5 BC EC Interpretation

Having reconstructed a statistically significant whole-brain effective connectivity using the Boost Connectivity (BC) method (Figure 5.6B), the next step is to interpret its organisation and functional implications. While the dense connectivity matrix provides a comprehensive representation of inter-regional interactions, meaningful structure can be obscured by its complexity. The following analyses therefore summarise the BC results to identify dominant patterns at both the modular and global levels, quantifying the direction, balance, and principal modes of effective interactions across the brain.

5.5.1 Module Summary

To further determine the influence of each functional module, the total 'flow' of brain activity incoming/outgoing and corresponding excitatory/inhibitory response of module is summarised. This allows for general comparison of the characteristics of each module, whether they are an activity sink or generator. The calculation for total connectivity that is incoming/outgoing & excitatory/inhibitory:

$$\begin{aligned}
 & \frac{1}{|\text{module}|} \sum_{i=0}^N \sum_T C_{iT} [C_{iT} > 0, T \in \text{module}] && \text{Incoming excitatory} \\
 & \frac{1}{|\text{module}|} \sum_{i=0}^N \sum_T C_{Ti} [C_{Ti} > 0, T \in \text{module}] && \text{Outgoing excitatory} \\
 & \frac{1}{|\text{module}|} \sum_{i=0}^N \sum_T C_{iT} [C_{iT} < 0, T \in \text{module}] && \text{Incoming inhibitory} \\
 & \frac{1}{|\text{module}|} \sum_{i=0}^N \sum_T C_{Ti} [C_{Ti} < 0, T \in \text{module}] && \text{Outgoing inhibitory}
 \end{aligned} \tag{5.3}$$

Where $|\text{module}|$ is the size of the module being examined. The module summary is calculated for the significant BC EC and the results are show in Figure 5.8. Figure 5.8 describes that across all modules, incoming excitation exceeds incoming inhibition (consistent with $\overline{C_{BC,ij}} > 0$), suggesting that inter-module exchange is predominantly excitatory. Another observation is that outgoing excitation is larger

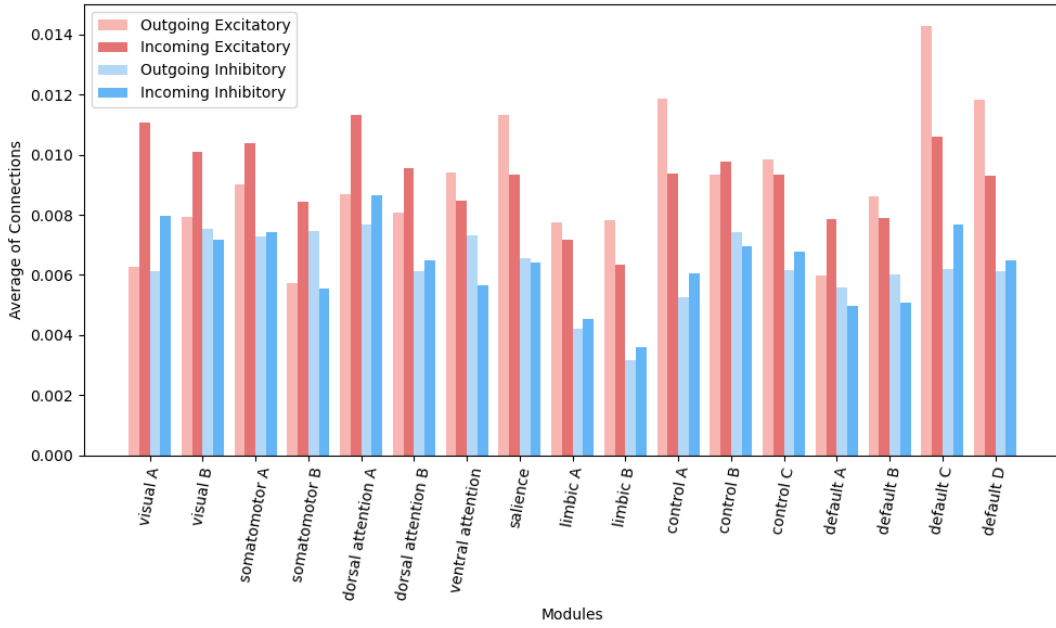


FIGURE 5.8. **BC EC Module summary characterises 'flow' and response to activity.**

Bar plot values are calculated according equations (5.3) and normalised by module size.

for almost all modules (e.g., dorsal attention, limbic, control, default), with the only exception being the somatomotor A module. If this result is valid this would suggest that the somatomotor module plays a significant role modulating the activity of other regions.

5.5.2 Principle Component Analysis

5.5.2.1 Method

The high dimensionality of the ECs obscure the meaningful global structure, since the fine-grained details of individual connections can dominate the visual representation. To extract the dominant patterns of interaction and summarise how regions co-vary in their causal influence, a dimensionality-reduction method such as Principal Component Analysis (PCA) can be applied. This enables the complex connectivity structure to be represented in a small number of components that capture the main sources of variance in the network.

Principal Component Analysis (PCA) is a linear orthogonal transformation that projects data onto a new coordinate system defined by directions of maximal variance (Jolliffe, 2002; Jolliffe and Cadima, 2016). Given a data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ with mean-centred variables, the covariance matrix is defined as

$$\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}. \quad (5.4)$$

The eigen-decomposition of $\mathbf{S}$,

$$\mathbf{S} = \mathbf{W} \mathbf{\Lambda} \mathbf{W}^\top, \quad (5.5)$$

yields orthogonal eigenvectors $\mathbf{W}$ (the principal component directions) and eigenvalues $\mathbf{\Lambda}$ that indicate the proportion of variance explained by each component. The first principal component corresponds to the linear direction that captures the maximum variance in the data, while each successive component captures the largest remaining variance under the constraint of orthogonality to previous ones.

5.5.2.2 Principle Components of EC

When PCA is applied directly to a single EC matrix, $\mathbf{C}$, it decomposes the spatial structure of connectivity into a set of orthogonal modes. In this case, each principal component identifies a pattern of directed connectivity (i.e., a weighted combination of source and target regions) that explains a portion of the variance across all connections in the network.

Each principal component can be reshaped back into matrix form. The first component often captures a global connectivity mode (e.g., overall excitatory coupling), while subsequent components may represent more specific functional patterns. Retaining only the top few components isolates the key modes of network organisation while filtering out noise and fine-scale fluctuations.

(Leonardi et al., 2013) demonstrated PCA for extracting dominant modes of functional connectivity from fMRI data, describing these as “eigenconnectivities.” Similarly, Vos de Wael et al. (2020) applied PCA to functional and structural connectivity matrices to uncover spatial gradients of cortical organisation, illustrating how the principal components capture large-scale, interpretable modes of brain network topology.

5.5.2.3 BC Components

PCA is applied to the BC EC results post identification of significant connections. By selecting the top 10 principle components, which explain the maximum amount of spatial variance and structure, then recombining to get a reduced EC. The top 3 components are also plotted alone, to identify directed functional modules. These results are in Figure 5.9.

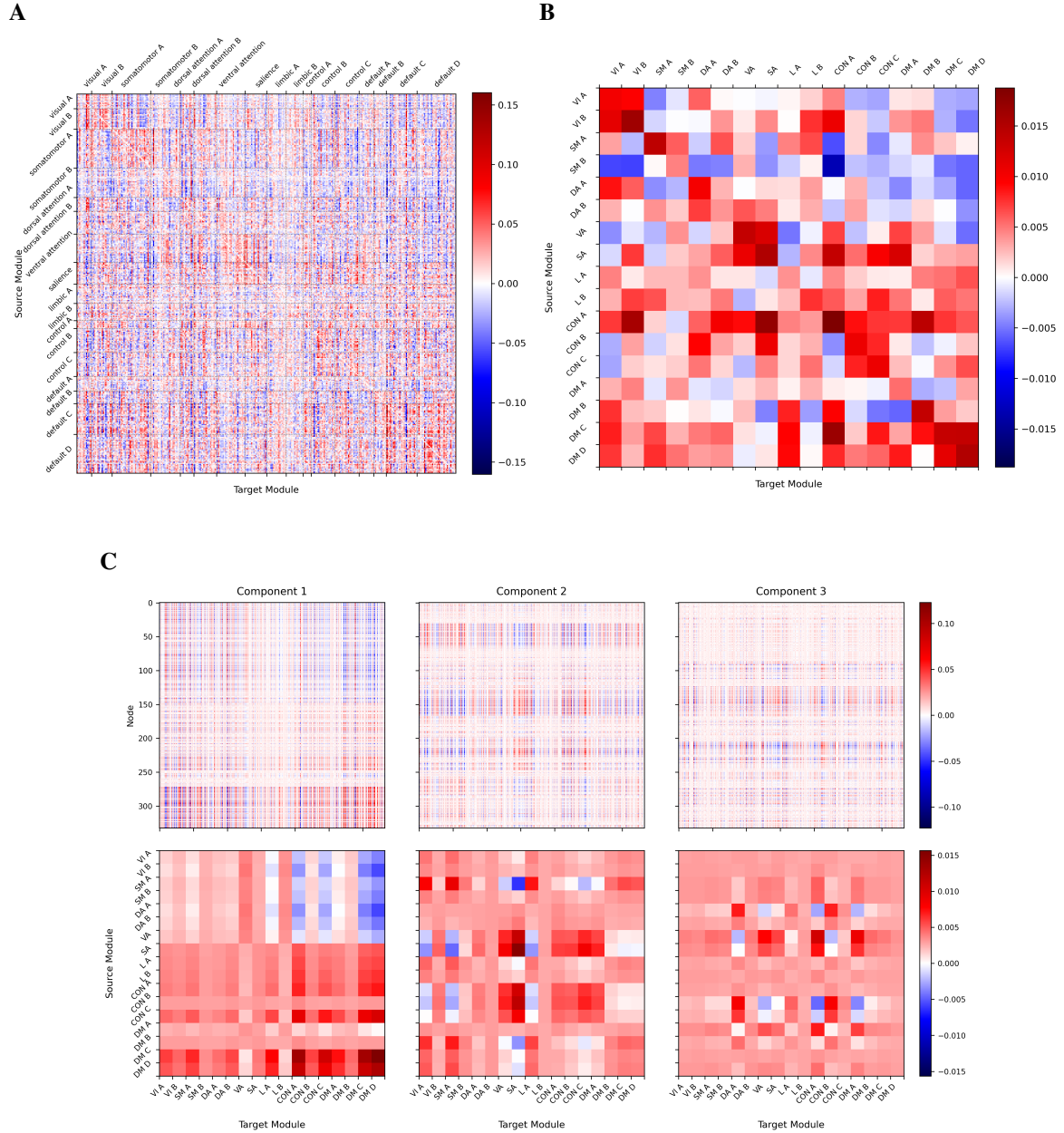


FIGURE 5.9. **PCA reduced BC EC & functional module average.**

(A) Reduced reconstruction of BC EC from top 10 principal components (explaining 17.4% of total variance). (B) Yeo's Module-level summary of PCA reduced BC EC. (C) Top 3 principal components identified by PCA reduction, which individually explain variance (left to right): 3.9×10^{-2} , 2.6×10^{-2} , 1.9×10^{-2} , with mean connectivities of: $|\overline{C}_1| = 8.7 \times 10^{-3}$, $|\overline{C}_2| = 7.1 \times 10^{-3}$, $|\overline{C}_3| = 5.9 \times 10^{-3}$.

The effect of PCA reduction on the BC is described Figure 5.9A and shows visual introduction of a global structure, such as the low modality limbic networks A & B which is consistent to the results

found in BA (Figure 5.6A). The module averaging then further summarises this structure (Figure 5.9B), highlighting features such as: inhibitory effects of visual, somatomotor & dorsal attention on the control & default mode networks, a strong intra-module excitatory relationship along the diagonal.

From Figure 5.9C, Component 1 again describes the inhibitory nature of the visual, somatomotor, dorsal attention & ventral attention on the control A, C and default C, D network. Component 2, contains strong intra and inter-excitatory interaction between the hyper modules of ventral attention & salience with the control A, B & C networks. Finally component 3, contains an even more complex network of interactions between ventral attention, salience, control B & C. A more detailed analysis of these components is contained within section 5.7.3

These components provide a compact representation of the brain’s causal architecture, revealing the principal organisational axes embedded within the dense EC matrix.

5.6 Delay Modelling

So far the model of brain activity using rsfMRI observations has relied on the instantaneous coupling from time $t - 1$. However, a growing body of work shows that the propagation lag of infra-slow fluctuations (typically < 0.1 Hz) vary across brain region pairs (Mitra et al., 2014, 2015a). Raut et al. (2019) argues for in the case of rsfMRI that temporal delays among regional rs-fMRI signals are “complementary” to zero-lag correlation for capturing the full spatiotemporal structure of brain activity. The observed zero-lag temporal results modelled thus far might be the result of a higher dimensional sequence of events over time scales ≈ 1 s (Mitra et al., 2015b). By explicitly modelling delayed connectivity, one can capture the influence of prior states over time ranges greater than the sample rate, 0.72s therefore may improve predictive performance in this infra-slow regime.

5.6.1 Delayed Network Model

The VAR model of brain described in equation 4.3, may be extended to consider time-lagged activity and a delayed connectivity:

$$\mathbf{x}(t+1) = \mathbf{x}(t)\mathbf{C}_0 + \mathbf{x}(t-1)\mathbf{C}_1 + \sigma \quad (5.6)$$

Where $\mathbf{C}_0$ is the canonical zero-lag matrix, representing instantaneous updates from the previous time step and $\mathbf{C}_1$ is the one time-lag matrix propagating updates from the activity two steps ago, σ is the representative of the random error in activity.

The network inference procedure can be extended to not only optimise for $\mathbf{C}_0$ but for $\mathbf{C}_1$ simultaneously, by concatenating the activity matrix with a delayed copy, $\mathbf{X}_{D=\{0,1\}}$ (given by equation 5.7).

$$\mathbf{X}_{D=\{0,1\}} = \begin{bmatrix} X_1 \\ X_0 \end{bmatrix}, \quad X_{D=\{0,1\}} \in \mathbb{R}^{(2N) \times (M-1)}.$$

$$\mathbf{X}_1 = \{\mathbf{x}(1), \mathbf{x}(2), \dots, \mathbf{x}(M)\}, \quad \mathbf{X}_0 = \{\mathbf{x}(0), \mathbf{x}(1), \dots, \mathbf{x}(M-1)\} \quad (5.7)$$

Applying the LLS regression (equation 4.6) reconstruction technique to $\mathbf{X}_{D=\{0,1\}}$, fits the network model for both X_0 and X_1 concurrently. The result is a block matrix $\mathbf{C}_{D=\{0,1\}} \in \mathbb{R}^{2N \times 2N}$, where each block represents:

$$\mathbf{C}_{D=\{0,1\}} = \begin{bmatrix} \check{\mathbf{C}}_0 & \check{\mathbf{C}}_1 \\ \mathbf{C}_1 & \mathbf{C}_0 \end{bmatrix}, \quad (5.8)$$

In which either $\check{\mathbf{C}}_0, \check{\mathbf{C}}_1$ or $\mathbf{C}_0, \mathbf{C}_1$ contains the correct connectivity and the other pair contains trivial connectivity as a result of being degenerate. This procedure can be applied to reconstruct arbitrary combinations of delayed connections (e.g. $\mathbf{C}_{D=\{0,10\}}$ or $\mathbf{C}_{D=\{0,1,2,3\}}$).

5.6.2 Results

Unfortunately the boost connectivity cannot be applied to the delay model framework, as the inclusion of the second $\mathbf{C}_1$ doubles the number of variables to optimise for. This further reduces the ratio of known to unknown variables, in which case already only produces partial ECs. Therefore the delay framework is applied to the boosted activity reconstruction to determine connectivities of $\mathbf{C}_0$ and $\mathbf{C}_1$ and results are described in Figure 5.10.

The striking feature between $\mathbf{C}_0$ and $\mathbf{C}_1$ is the alternating excitatory and inhibitory nature. That is for a region that receives an instantaneous excitatory response, it then receives from an inhibitory response a step later from the same region.

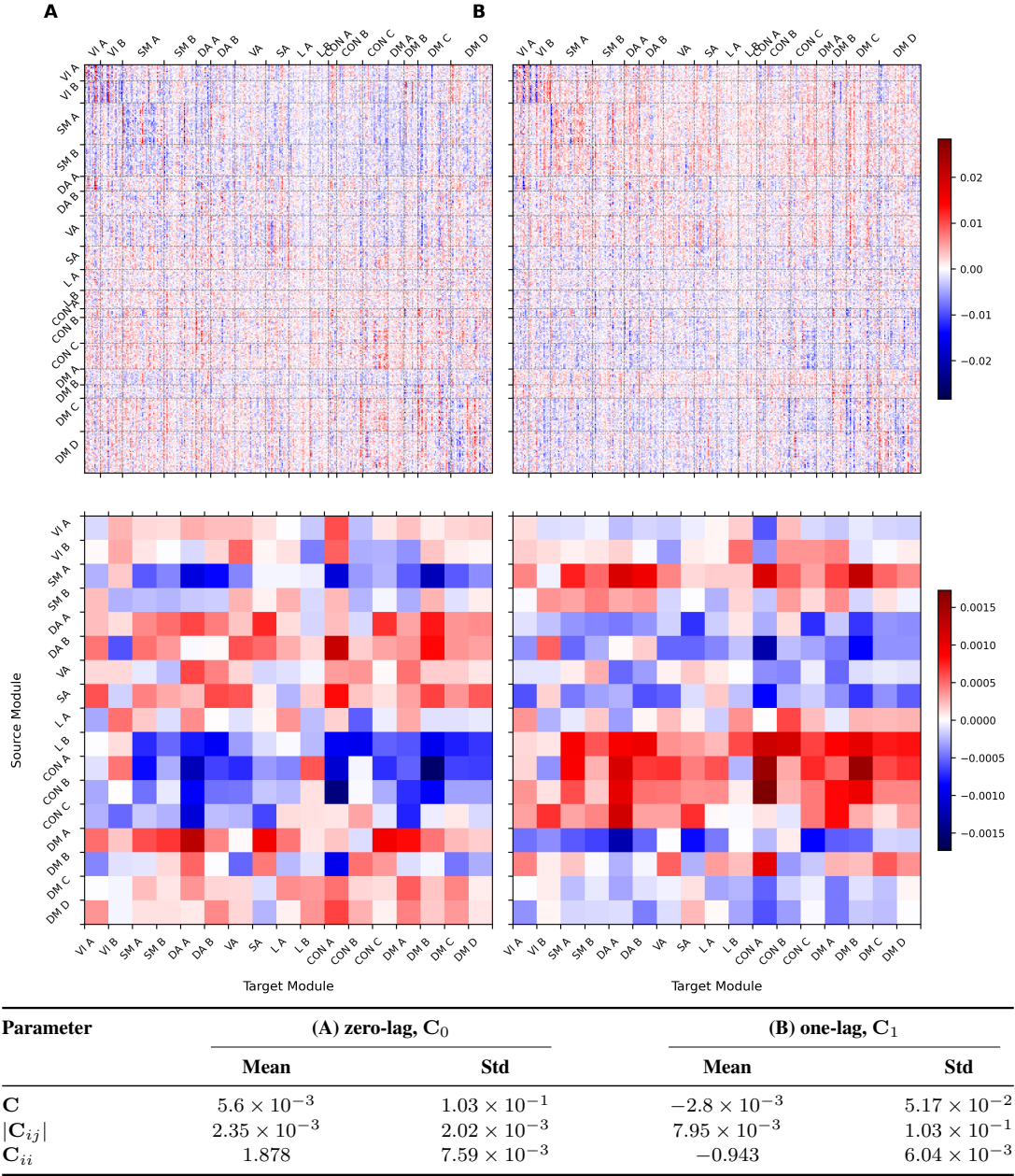


FIGURE 5.10. **Reconstructed BA delay connectivities, $D = \{0, 1\}$.**
 (A) Zero time-lag connectivity, C_0 , with module averages. (B) One time-lag connectivity, C_1 , with module averages.

Visually there is a strong horizontal alignment between between the region responses (excitatory/inhibitor) and the modules borders, suggesting a good fit with Yeo's functional modules.

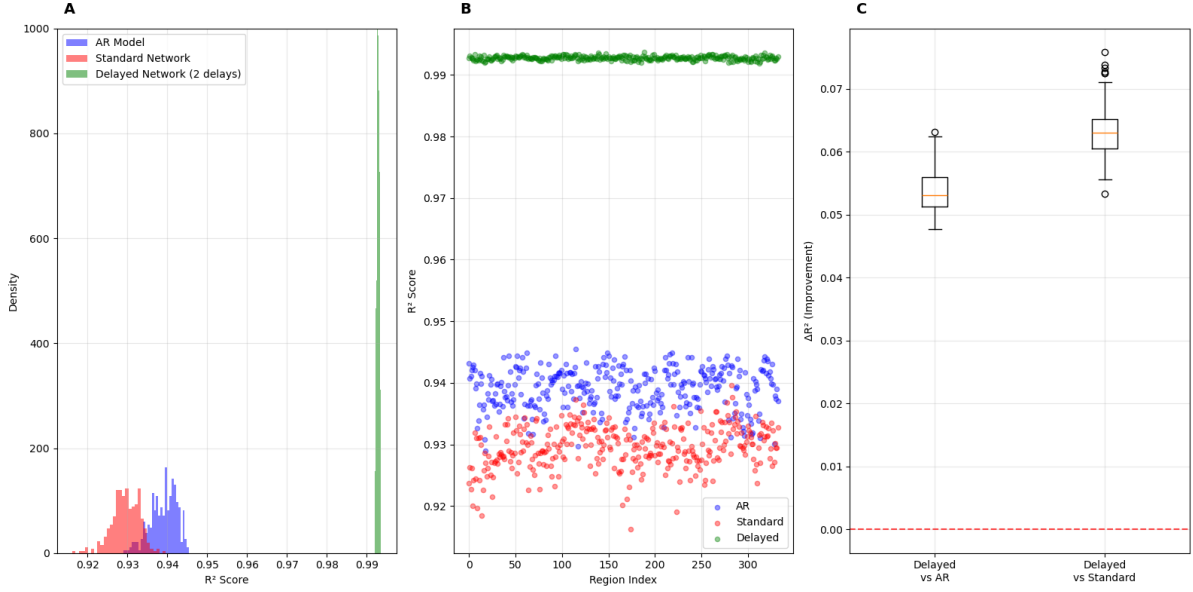


FIGURE 5.11. Proportion of brain activity variance explained by the 'Delayed Network' model's predictions.

(A) Frequency density of regions R^2 scores for each model: delayed (Green), network (Red), AR (Blue). The average R^2 scores; $\overline{R^2}_{\text{delay}} = 0.99 \pm 0.0003$, $\overline{R^2}_{\text{network}} = 0.93 \pm 0.0034$, $\overline{R^2}_{\text{AR}} = 0.939 \pm 0.0033$. (B) Scatter of each regions R^2 score. (C) ΔR^2 improvement of delay model over alternatives: $\Delta R^2_{\text{network}} = 0.063 (p \ll 0.05)$, $\Delta R^2_{\text{AR}} = 0.054 (p \ll 0.05)$

5.6.3 Delayed Network Model: Goodness of fit

Using the 'Delayed Network' model description from equation 5.6 and the reconstructed zero/one time-lag EC displayed in Figure 5.10 the the goodness of model fit was determined similarly to section 5.4.3 by quantifying the proportion of variation in activity explained by the models predictions (equation 5.2). By comparing this to the 'Standard Network' of zero-lag and the auto-regressive model, any improvements in model description can be determined. The results are in Figure 5.11.

As evident in Figure 5.11, the predictive ability of the delayed model is superior to both the standard and the auto regressive model, capable of predicting $\approx 100\%$ of the variation in activity. The significant improvement in the fit of the model indicates that the access to two time steps provides critical information for predictions.

5.6.4 Static/Derivative Model Interpretation

While the delayed network model demonstrates strong predictive power, interpreting the combined influence of two temporally separated connectivity matrices, $\mathbf{C}_0$ and $\mathbf{C}_1$, is not straightforward. The emergence of alternating regional responses across time lags is unlikely to reflect true structural connections at distinct temporal offsets; rather, it may arise from the model implicitly capturing response changes due to recent changes activity. To clarify this effect, a reformulation is proposed—the static/derivative model, which separates the instantaneous and temporal-change contributions to the dynamics, as described by the following equation:

$$\begin{aligned}
 \mathbf{x}(t+1) &= \mathbf{x}(t)\mathbf{C}_0 + \mathbf{x}(t-1)\mathbf{C}_1 \quad \text{let } \mathbf{C}_0 = \mathbf{D} + \tilde{\mathbf{C}}/2, \quad \text{and } \mathbf{C}_1 = \tilde{\mathbf{C}}/2 - \mathbf{D} \\
 &= \mathbf{x}(t)(\mathbf{D} + \tilde{\mathbf{C}}/2) + \mathbf{x}(t-1)(\tilde{\mathbf{C}}/2 - \mathbf{D}) \\
 &= \left[\frac{\mathbf{x}(t) + \mathbf{x}(t-1)}{2} \right] \tilde{\mathbf{C}} + \left[\frac{\mathbf{x}(t) - \mathbf{x}(t-1)}{1} \right] \mathbf{D} \\
 &= \mathbf{x}(t - \frac{1}{2})\tilde{\mathbf{C}} + \frac{\Delta \mathbf{x}}{\Delta t} \mathbf{D}
 \end{aligned} \tag{5.9}$$

Where $\mathbf{D} \in \mathbb{R}^{N \times N}$, represents the each regions response to change in activity, $\Delta \mathbf{x} / \Delta t$ ($\Delta t = 1$ for $D = \{0, 1\}$). $\tilde{\mathbf{C}}$ is the response to the average static activity at a time $\tilde{\mathbf{x}}(t - 1/2)$, assuming an average activity between the last two observations exists. Solving for $\mathbf{D}$ and $\tilde{\mathbf{C}}$ results in:

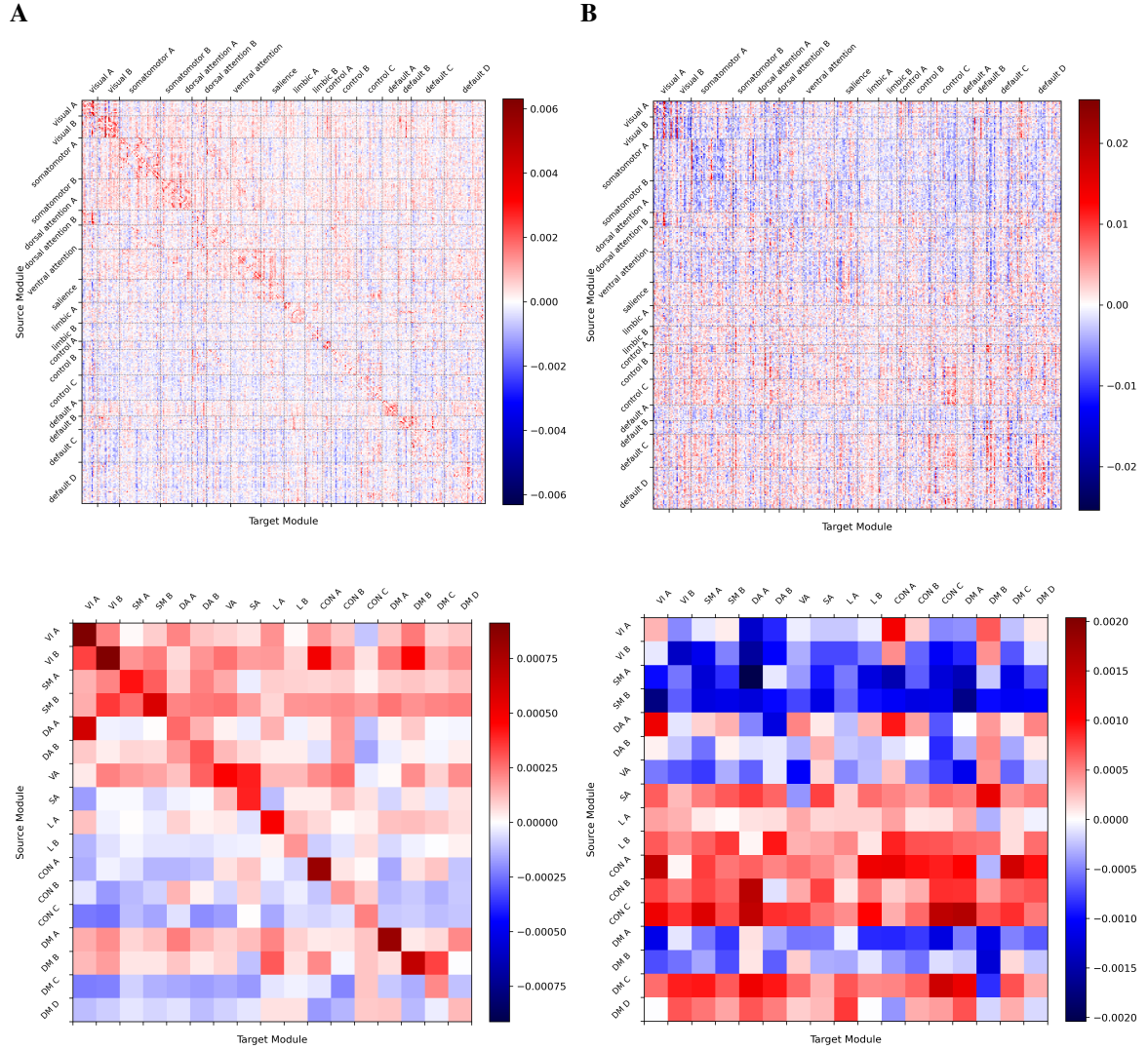
$$\mathbf{D} = \frac{\mathbf{C}_0 - \mathbf{C}_1}{2} \quad \tilde{\mathbf{C}} = \mathbf{C}_0 + \mathbf{C}_1 \tag{5.10}$$

Both $\mathbf{D}$, the 'derivative' and $\tilde{\mathbf{C}}$, static 'connectivity' are are plotted in Figure 5.12.

The static EC resembles the original BA EC (Figure 5.4B), with excitatory projections from visual, somatomotor, dorsal attention, and default A/B, and largely inhibitory projections from limbic, control, and default C/D. Module averages again show strong intra-module excitation (e.g., visual and default A/B).

The derivative EC displays the opposing response to the static delay EC, for when a region is activated by another (Static EC), the region is simultaneously inhibited by increases in the others activity (Derivative EC). The module average of the derivative EC does not display any intra-module structure along the diagonal, which prevents it from runaway activity growth.

To compare the relative contributions of $\mathbf{D}$ and $\tilde{\mathbf{C}}$, both the average magnitudes of the activity and its temporal change must be taken into account. The effective influence of the static component is given



Parameter	Static EC		Derivative EC	
	Mean	Std	Mean	Std
$\mathbf{C}$	2.87×10^{-3}	5.12×10^{-2}	4.20×10^{-3}	7.72×10^{-2}
$ \mathbf{C}_{ij} $	3.88×10^{-4}	3.56×10^{-4}	2.31×10^{-3}	1.97×10^{-3}
$\mathbf{C}_{ii}$	0.935	3.34×10^{-3}	1.411	6.65×10^{-3}

FIGURE 5.12. Static and Derivative EC (from BA).

(A) Static EC, $\tilde{\mathbf{C}}$ from delay modelling, including Yeo's Modules averages (B) Derivative EC, $\tilde{\mathbf{D}}$ from delay modelling, including Yeo Modules averages.

by $\langle \overline{\mathbf{x}(t)} \rangle \times |\tilde{\mathbf{C}}| = (1.51 \pm 0.24) \times 10^{-4}$, while that of the derivative component is $\langle \overline{\Delta \mathbf{x} / \Delta t} \rangle \times |\tilde{\mathbf{D}}| = (7.5 \pm 0.89) \times 10^{-5}$. On average, the static activity at time $t - \frac{1}{2}$ therefore exerts a stronger influence on future activity than the recent change in activity, approximately by an order of magnitude.

5.6.4.1 Discussion

Now to understand the effect the alternative excitatory/ inhibitory response between the $\tilde{\mathbf{C}}$ and $\mathbf{D}$, firstly consider a relationship representative of equation 5.9: for a single interaction between the activities of module x and y :

$$x(t+1) = c \cdot y(t) + -d \cdot \frac{dy}{dt}(t), \quad c, d \in \mathbb{R}, c > 0, d > 0 \quad (5.11)$$

$$\text{If you consider a general periodic function: } y = \cos(t) \quad (5.12)$$

$$x(t+1) = c \cdot \cos(t) + d \cdot \sin(t) \quad (5.13)$$

The resulting contribution to module x from module y is one part in-phase: $c \cdot \cos(t)$ and one part out of phase by 90 degrees: $d \cdot \sin(t) = d \cdot \cos(t - \pi/2)$. In this model: c can be viewed as the amount of co-activation and d can then be viewed as the amount of lag in the response of x to changes in activity of module y . The contribution of the correlated $c \cdot y(t)$ and the lagged $d \cdot \cos(t - \pi/2)$ is based on the relative ratio between $c : d$. For the case where $c < 0, d < 0$, then resulting contribution of y is a lagged anti-coactivation response.

Relating back to the static delay EC, $\tilde{\mathbf{C}}$ and the derivative EC, $\mathbf{D}$ you see that a regions response is the combination of it's co-activation and lags to each other region:

$$x_i(t+1) = \sum_{j=0}^{333} \left[\tilde{x}_j(t - \frac{1}{2}) \tilde{C}_{ji} + \frac{\Delta x_j}{\Delta t} D_{ji} \right]$$

As calculated previously, the static effective connectivity exerts approximately ten times greater influence on activity than the derivative term when accounting for the relative magnitudes of activity and its rate of change. This implies that the lagged response corresponds to roughly $\frac{\pi/2}{10} \approx \pi/20$ of a phase on average. It should be noted, however, that this is a simplified illustration, the true dynamics are considerably more complex once reciprocal inter-regional interactions are taken into account.

5.6.5 Static Delay Network PCA

To complete the interpretation of the static/derivative network, PCA is applied to the static EC to determine the dominant spatial components. The results are in Figure 5.13.

Figure 5.13B indicates strong excitatory relationships within module pairs (e.g., visual A/B, somatomotor A/B, dorsal attention A/B; ventral attention with salience). In Figure 5.13C, Component 1 separates

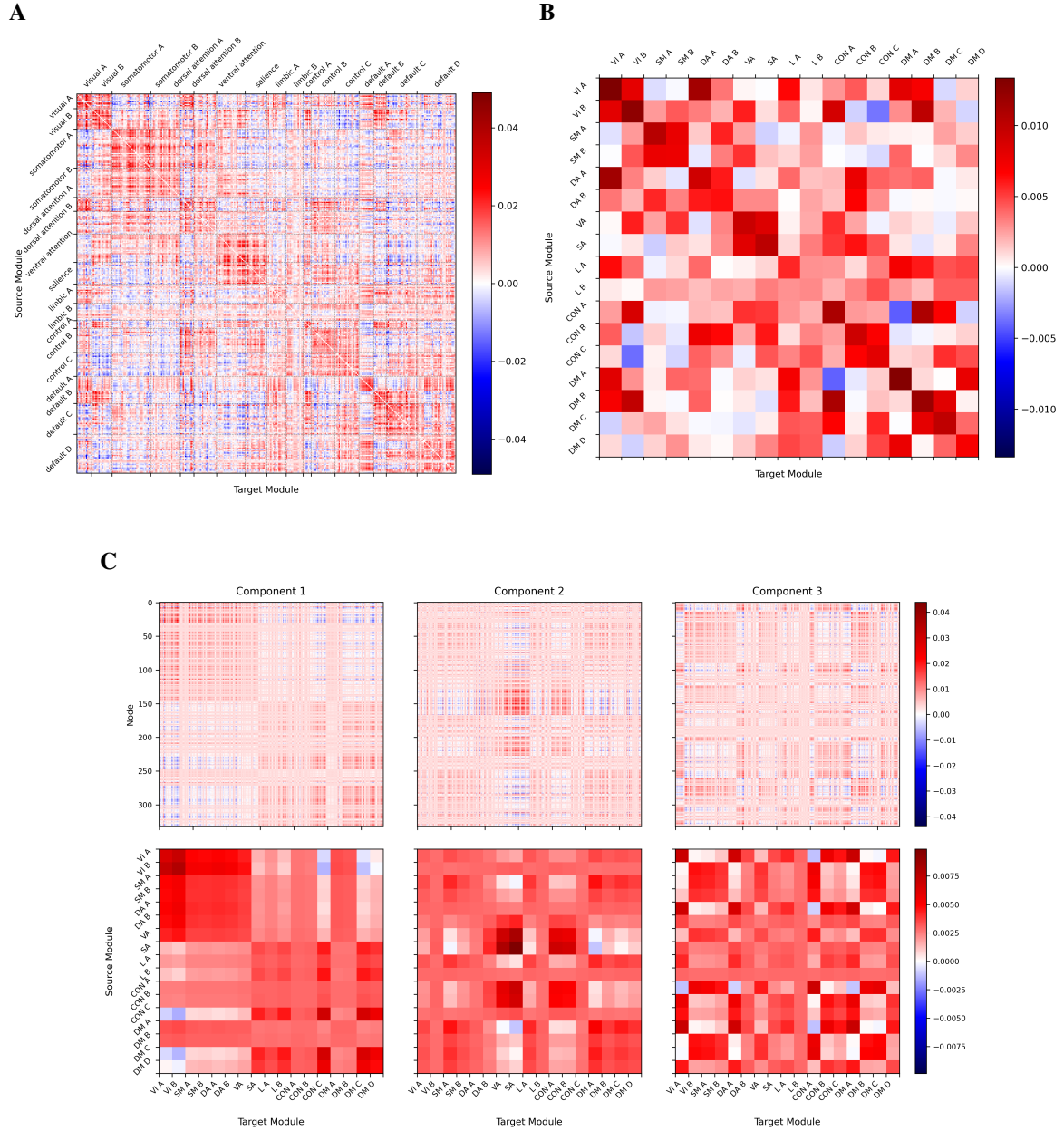


FIGURE 5.13. **PCA reduced static EC & Yeo's functional module average.**

(A) PCA reduction of static connectivity from top 10 principal components (explaining 3.2% of total variance). (B) Yeo's Module-level summary. (C) Top 3 principal components identified by PCA reduction, which individually explain variance (left to right): 3.3×10^{-3} , 3.2×10^{-3} , 3.2×10^{-3} , with mean connectivities of: $|\overline{C_1}| = 3.4 \times 10^{-3}$, $|\overline{C_2}| = 3.4 \times 10^{-3}$, $|\overline{C_3}| = 3.4 \times 10^{-3}$.

two broad groups—sensory/attention (visual, somatomotor, dorsal and ventral attention) versus control/affect (limbic, salience, control C, default C/D). Component 2 emphasises an excitatory subnetwork comprising ventral attention, salience, and control A/B; Component 3 exhibits a distinct, regular motif.

The dimensional reduction of the static delay EC by selection of the top 10 principal components produced a nearly symmetric EC (Figure 5.13B), which is in contrast to the result of the PCA BC which remained asymmetric (Figure 5.9A).

The process of applying PCA to the entire connectivity results in a symmetrical result only if the first selected principal components are themselves symmetric, implying that that EC principal axes of variance lie largely in a symmetric subspace. In other words the strongest structural relationships inferred by BA EC (including the delay model) are are bidirectional, undirected couplings.

Whilst this challenges the entire purpose of the directional extension to functional connectivity that linear least squares regression provides, it provides further insight into the explanation that BA EC provides. In conjunction with the statistical definition of the boosting activity process: these results imply that the most influential of the ubiquitous features which generate brain activity is symmetric couplings.

5.7 Discussion

5.7.0.1 Inter-Subject Variability

In an attempt to skirt the incomplete reconstructions for single-subject EC's the fMRI activity was concatenated to artificially increase the sample size and improve reconstruction, a technique referred to as Boost Activity (BA) throughout the results. LLS regressions assumes that temporal noise be normally and independently distributed about the true value. An implication of this is that the variability amongst the subject's concatenated activity must also be independently distributed.

Whilst knowledge on the directed effective connectivity is limited, there is growing body of evidence shows that functional connectivity (FC) is not uniform across individuals, even when they undergo the same resting-state protocols (Laumann et al., 2015). For example Mueller et al. (2013) found that while sensorimotor regions exhibited relatively low inter-subject variance, both the ventral and dorsal attention displayed much higher variability. Li et al. (2021) goes as far to associate the inter-subject variability with underlying gene expression and structural differences, using similar HCP fMRI data. These studies support the conclusion that an individual's FC patterns are meaningful and structured, not simply noise, and that concatenating many subjects into a "group" would violate spatial temporal independence.

When each subject contributes a distinct connectivity architecture, aggregating their time series reduces the signal of any one subject's network while inflating residual variability. Thus, the dramatic reduction in significant edges under BA (Figure 5.6) aligns with theoretical expectations.

5.7.0.2 Group Averaging

To form a robust group average from the incomplete single-subject EC, the most reliable method was the boosting connectivity (Figure 5.4A). Under the assumption that the noise introduced from the partial reconstructions is independent averaging would produce a BC EC which would be the average sample drawn from the population of subjects. Generation of surrogate time-series striped of casual interactions revealed that $\approx 86\%$ of connections were arose from significant interactions between regions, which could not be explained by the null model generated from the surrogate time series. Unfortunately it is not possible to determine the proportion of activity explained by the BC EC model as demonstrated in Figure 5.7, due to the inability to produce single-subject EC for individual activity predictions.

5.7.1 BC BA Final Comparison

This chapter has extensively compared the advantages of both the Boost Activity (Fixed Effect) model and the Boost Connectivity (Random Effect) model. In theory the BA EC describes the only connections of brain which are the strongest/consistent influence on the generation of activity across all subjects. While the BC EC is a the average connectivity drawn from the population of partial subject connectivities.

The BA EC was a found to be limited in several facets: few inter regional connections were found to be statistically significant, insignificant ability to predict activity indicating improper model fit and general violation of the LLS assumption of temporally independently noise.

The BC effective connectivity reconstruction demonstrated robustness when compared to a null model constructed from surrogate time series and revealed a rich set of directed functional components, several of which aligned with established findings in the literature.

Comparing the BA and BC models offers valuable insight into the consistency of network features across individuals, particularly in data-limited regimes. However, given the methodological advantages and the robustness discussed above, the group-level Boost Connectivity (BC) effective connectivity is the stronger candidate for representing the whole-brain effective structure

5.7.2 Alignment with Literature

There were several derived results from group representative BC EC that align with current neuroscience literature. These conclusions might not be novel but bolster the validity further inferences.

5.7.2.1 Homogeneous Functional Modules

There are several examples of strong alignment between the ECs response behaviour (excitatory/inhibitory) and superimposed Yeo's functional modules, such as Figures 5.12&5.9. This suggests a consistency between the inferred directed connections and inferred functional segregation determined in Yeo et al. (2011). Future work should include a method to quantitatively determine the alignment with a given parcellation/module scheme.

5.7.2.2 Net Excitatory

A consistent result for both the BA and BC EC networks was the net excitatory nature, $\overline{C} > 0$ (e.g. $|\overline{C}_{BC}| = 2.9 \times 10^{-3}$, Figure 5.4A), clearly demonstrated by the greater outgoing excitatory of each module in the activity flow chart (Figure 5.8).

Literature suggests that at the inter-regional scale, long-range projections are known to be predominately glutamatergic (primary excitatory neurotransmitter) and generate mostly excitatory stimulation to their target regions (Brown and Hestrin, 2009; Leong et al., 2016). Whilst inhibitory connections contribute locally, sculpting the gain within regions. Both the processes of fMRI parcellation into the 333 regions and then module averages would suppress the expression of inhibitory connections amongst the long distance excitatory connections. The remaining inhibitory connections are representative of larger system wide opposing networks. Hence there is a good consensus between the current literature and the results demonstrated.

5.7.2.3 Inhibition of Default Mode networks

Results from the principal component analysis, component 1 from Figure 5.9C & 5.14, describe the inhibition of higher-order brain regions such as the default and control networks by the lower-order visual and dorsal-attention networks. This result aligns with current literature: for example, "Top-down

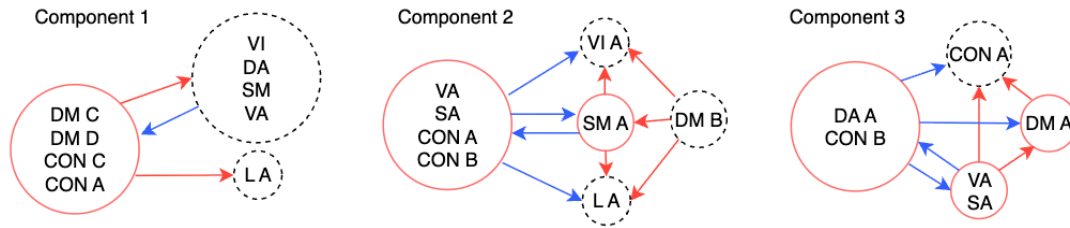


FIGURE 5.14. **Overview of of directed functional module interactions for BC EC PCA.**

The overview is derived from components from the BC EC PCA as described in Figure 5.9C. Groups are formed by combining modules which have similar inwards and outwards going connections. Red connections indicate excitatory connections, Blue indicates inhibitory connections. Red border indicates modules/groups which excite within the module. Dotted border indicates a group with no clear internal connections. Components were selected according to greatest variance explained.

regulation of default mode activity in spatial visual attention" from Wen et al. (2013) found that during visual spatial-attention tasks, top-down signals from task-control networks exert inhibitory influence on the Default Mode Network (DMN). Similarly Hellyer et al. (2014) reported that activation of the control/dorsal-attention networks is linked to suppression of DMN activity and decreased global network metastability, consistent with a dominance of externally-directed attention systems over internal ones.

5.7.3 Network Discoveries

The BC EC describes the group representation of the 471 subjects within the HCP rs-fMRI study. Through module averaging and PCA several systems of directed networks were identified, which are described in Figure 5.14.

Component 1 reflects a clear segregation between lower-order networks, which process external sensory and motor information (visual and somatomotor systems) and guide both voluntary and involuntary attention (dorsal and ventral attention networks), and higher-order networks involved in internal regulation, such as cognitive control, memory, and emotion (control and default-mode networks). The higher-order networks exhibit strong intra-network excitatory coupling, supporting sustained internal activation and integration of self-referential processes, whereas the lower-order systems appear to exert an inhibitory or modulatory influence, constraining this activity in response to external demands. This component is consistent with the first component identified by application of PCA to the static EC (Figure 5.13C, displaying strong high/lower network order division which inhibit each other.

Component 2 describes one major system comprised of the ventral attention, salience and control A & B networks, all of which excite one another. Combining the attributed functions of the individual networks, the system is theorised to drive the switch from external stimulus to a cognitive response. This system then competes with both visual and somatomotor networks, possibly the originators of the stimulus by mutual inhibition. The default mode network B passively oppose the larger system by exciting the visual, somatomotor and limbic networks whilst having no incoming connections. Again this component shares similarities to the second component identified by the static EC PCA, a system of ventral and salience opposing the visual and somatomotor (Figure 5.13C).

Component 3 captures a competitive dynamic between internally driven, goal-directed networks (dorsal attention and control B) and externally oriented attention systems (ventral attention and salience). Beyond this broad interpretation, the interactions become more difficult to explain: control B appears to inhibit control A, a network of similar functional role. To add external stimuli processed by the ventral attention network seem to excite the default mode network, typically associated with introspective or mind-wandering states. These counterintuitive relationships suggest that the underlying interactions are more complex than the linear framework can fully capture, potentially reflecting higher-order or context-dependent dynamics not represented in this model.

5.8 Summary

The analyses conducted on the HCP dataset reveal several key findings. First, the rsfMRI recordings are dominated by infra-slow fluctuations (<0.1 Hz), confirming that the measured activity evolves at a rate much slower than the sampling interval. Despite these slow dynamics, the dataset reproduces a clear and structured functional connectivity pattern that aligns closely with Yeo's functional modules, validating both the preprocessing pipeline and the expected large-scale organisation of resting-state networks.

However, the sluggish temporal evolution of the signals limits the effective number of independent samples, making single-subject reconstructions of effective connectivity unreliable. To overcome this, two strategies were considered for generating a group-level estimate: boosting connectivity (BC) and boosting activity (BA).

While BA theoretically increases sample size, it proved highly sensitive to inter-subject variability, producing reconstructions dominated by intrinsic self-connections . In contrast, BC preserved meaningful between-region structure by averaging subject-level reconstructions, yielding a more interpretable population-level network.

Model-based activity prediction confirmed the limited explanatory power of BA, as the reconstructed connectivity failed to improve predictive accuracy beyond autoregressive baselines. Incorporating temporal delays, however, substantially improved model fit.

Principal Component Analysis (PCA) applied to the BC effective connectivity further revealed coherent, large-scale patterns of interaction, identifying dominant modes that correspond to known functional systems.

The introduction of one time-lag connections into the network model significantly improved the predictability and fit of the 'delayed network' model. This suggests in the infra-slow regime, the activity readings beyond a single time sample provide nuance and necessary information.

To aid interpretation of the temporally separated connectivity terms, a static/derivative framework was developed. This formulation provides a clearer description of lagged responses between regions by distinguishing between the influence of current activity and its recent rate of change.

Finally, several functional components identified within the directed BC effective connectivity showed strong correspondence with established findings in the literature. This consistency lends confidence to the reconstruction, allowing the discovered directed network components to be further interpreted in terms of their potential biological functions.

CHAPTER 6

Conclusion

Before concluding this report, it is important to acknowledge the key limitations and potential threats to the validity of the findings. These factors have been carefully considered throughout the study's design and analysis, and deliberate efforts were made to mitigate their impact wherever possible. Nonetheless, recognising these constraints provides essential context for interpreting the results and defining the scope of the conclusions drawn.

6.1 Limitations

There are several limitations with the work presented in the report. Some limitations stem from underlying assumptions present in current literature and some originate from limitations in the method and some from access to resources.

6.1.1 Sources of Noise

With fMRI measurements there are several sources of noise to account for. Head motion, physiological fluctuations (cardiac pulsations and respiratory cycles) and thermal/electronic noise all independently influence readings and the downstream analysis. "Physiological and motion signatures in static and time-varying functional connectivity and their subject identifiability" by (Xifra-Porxas et al., 2021), discuss in detail the sources of fMRI artifacts and the influence on identifying individual subjects: motion leads to mismatched voxel (region) positions inducing spurious correlations between possibly non-interacting regions, changes in cardiac and respiratory cycles alter the cerebral blood flow yield BOLD signal oscillations and thermal fluctuations whilst generally Gaussian distributed sets a fundamental limit on fMRI sensitivity (the noise floor). Bright and Murphy (2015) applied independent component analysis to the sections of the fMRI data built by regressors related to head motion and physiological variability and demonstrated characteristic structures similar to the resting-state network, showing the artifacts ability

to confound analysis, a result corroborated by Xifra-Porxas et al. (2021). Though with proper identification of the slow-symmetric oscillations associated with noisy artifacts the "aggressive" preprocessing strategies can mitigate the influence (Xifra-Porxas et al., 2021).

6.1.2 Underlying Assumptions

Stationarity Assumption. A popular assumption amongst model-based approaches of brain connectivity, including linear least squares is the existence of a time-invariant connectivity (Gilson et al., 2018; Li et al., 2019; Khanjaniapour et al., 2024). This stationarity assumption implies that the relationships between regions of the brain do not change over time, and that a singular connectivity structure explains the activity over the entire recording. This assumption is convenient, improving mathematical tractability, providing more time points to reconstruct a single connectivity, easing comparability as there is only a single canonical connectivity.

Several studies have investigated non-stationarity in the human brain during fMRI and claim that the brain transitions between states such as the default mode into an active state during cognitive loads, violating the stationarity assumption (Hansen et al., 2015; Allen et al., 2014). Alternative approaches have been suggested such as producing a dynamic connectivity as described by Park et al. (2018), implementing a sliding window to infer connectivity of over a range of time. Not modelling for a dynamic connectivity may result in larger intra and inter-subject variability as several operating modes are averaged over time, obscuring an accurate individual representation.

6.1.2.1 Gaussian Noise

Earlier fMRI analyses assume/model the noise within activity measurements to be Gaussian or even "white" noise, meaning uncorrelated in time and normally distributed ("white" also implies a random signal evenly across all frequencies) (Wink and Roerdink, 2006). This assumption underlies several models and preprocessing steps including the maximum likelihood estimator for linear least squares, with the inclusion of an independent residuals.

Several studies in reality claim that the white-noise assumption is generally not strictly valid and that the noise present exhibits substantial temporal autocorrelation due to the structure noise sources as discussed earlier (Lund et al., 2006). The assumption of simply Gaussian (normal) distribution of fMRI

noise is usually better satisfied. MRI thermal noise is fundamentally Gaussian before image reconstruction, and after reconstruction it follows a Rician distribution that becomes nearly Gaussian at typical fMRI signal-to-noise levels (Wink and Roerdink, 2006). This means standard parametric statistical tests (which assume normality of errors) are usually robust for fMRI, as long as the data are preprocessed appropriately.

Failing to model the autocorrelation leads to invalid statistical inferences, p-values may be too low (finding significance too easily) because the model “thinks” there are more independent time points than there truly are (Lund et al., 2006)

6.1.3 Analysis

Information Reduction. With the advancements in fMRI technology whole brain scans produce high spatial resolute results, a wealth of information that allow in depth analysis. This study considers 333 regions, resulting in $333 \times 333 \approx 111,000$ connections. For example the delay EC, shown Figure 5.10, contains rich information regarding the inhibitory/excitatory strength of each of the 100’s of thousands of region pairs. The immense amount of information necessarily requires summarising to make practical inferences about the observed world.

The process of averaging over modules or separating out orthogonal principle component allows for further analysis but requires the removal of informative data. For example consider the region level diagram of the principle components in Figure 5.13C, there is clearly large regions of inhibitory connections which vanish once applying module averaging limiting further analysis.

6.1.4 Results

6.1.4.1 Single Subject EC

A clear limitation demonstrated is the inability to produce single-subject effective connectivities from the HCP rs-fMRI dataset (Figure 5.2). It was concluded that while there was in theory sufficient samples, slow fluctuations led to highly correlated dependent activity. Without reliable single-subject effective connectivities analysis regarding inter-subject variability: identifying modules which are consistent

across a population and classifying individuals based on their EC fingerprint is not possible. These processes are critical to further study into identifying how cognitive condition alter the connections within the brain, a powerful diagnostic tool.

6.2 Threats to Validity

Throughout the design of the method and the analysis of the results, critical decisions are made to present valid yet interpretable conclusions. It's worthwhile to consider the options and procedures considered for a balanced understanding of the validity of the study.

6.2.1 Internal Validity:

Is there any internal biases and confounding factors impacting the results.

6.2.1.1 Yeo's Functional Modules

As introduced in Chapter 5, Yeo's 7-network functional modules were used to group and reorder the 333 cortical parcels from Gordon's parcellation. This enabled both visual interpretation of intra- and inter-module connectivity patterns and the summarisation of module-level averages. The choice of module grouping has a non-trivial influence on both the quantitative results and qualitative interpretations.

Qualitatively, parcellation also shapes the interpretation of results. Using coarse modules like Yeo's emphasizes broad functional systems and can highlight large-scale organisation (e.g., default mode vs. control networks), but may mask finer distinctions within networks. Grouping high-resolution ROIs (e.g., Gordon parcels) into Yeo's modules enhances interpretability while smoothing over intra-module variability.

Yeo's parcellation (Yeo et al., 2011) was selected here due to its widespread adoption and empirical robustness in identifying replicable large-scale networks across individuals and datasets. It provides a canonical reference frame, making the results more interpretable and comparable to prior literature.

From the results presented in this report, there appears to be a strong correspondence between the derived whole-brain effective connectivity and Yeo's functional modules (Figure 5.9A & 5.10). This is evidenced by the clear clustering of stronger connections within modules and distinct boundaries between them, consistent with the modular structure hypothesised in the Yeo framework.

6.2.2 External Validity

How does the methods used extend to external research?

The LLS network inference technique has application beyond the HCP rs-fMRI dataset. There are only several features the brain activity requires to perform a stable group EC reconstruction:

- (1) Stable readings, activity is bounded
- (2) Largely independent noise (resting-state observations)
- (3) Sufficient independent samples, $M \gg N$

In the case for resting-state fMRI studies this often means applying several preprocessing steps outlined in the method but for alternative measures of brain activity would satisfy as long as they satisfy the assumptions of LLS.

6.2.3 Construct Validity

Construct validity concerns whether the chosen measures and methods are appropriate for addressing the research questions.

Several metrics were evaluated to quantify the effectiveness of network reconstruction, including the residual error, correlation, coefficient of determination (R^2), and the 'quality of reconstruction'. Preliminary experimentation revealed that the residual error was an unreliable measure for comparing networks with differing parameters, as variations in overall connection strength influenced both the absolute and percentage residuals. Similarly, correlation and R^2 tended to overestimate model performance because consistent over- or under-estimation of specific connection groups (intrinsic or extrinsic) could still yield high aggregate similarity.

The 'quality of reconstruction' metric, $Q_{0.95}$, as proposed by Shandilya and Timme (2011), was found to provide the most robust and interpretable assessment across varying network configurations. Unlike the other measures, $Q_{0.95}$ consistently reflected the proportion of correctly inferred connections and therefore offered a more reliable indicator of reconstruction accuracy.

6.2.4 Conclusion Validity

Are the results drawn from the data are credible and supported by appropriate statistical evidence?

Several steps were taken to strengthen confidence in the reported findings. For the synthetic network experiments, error bars representing one standard deviation across ten independent trials were included in all comparative plots, providing a measure of variability and reproducibility. In the construction of the whole-brain effective connectivity (BC EC), statistical filtering was applied to remove connections that were not significantly different from those in a null model lacking causal interactions between regions. This ensured that the retained connections were statistically justified and that the inferred EC more likely reflected genuine interactions among brain regions rather than random fluctuations or noise.

My academic background includes the study of Computer science, Data science and Physics which provides the foundational skills to quantitatively analyse the fMRI time series data and the ability to train and test models. On the other hand I have no higher training in biology and or neuroscience. This limits my ability to qualitatively make associations between neurological processes and the numeric results achieved in this report. Through my best effort and research I have made general claims regarding biological implications with the support of literature. Though this knowledge is introductory and further training would broaden the reach of the results.

To curb this inexperience, each of the results were related to existing literature where possible. Several findings were consistent with existing research of the fMRI analysis and whole brain connectivity, including: infra-slow fluctuations hinder the reconstruction effort, net excitatory nature of brain connections a long ranges and the opposing higher and lower order networks of the brain. Reconstructing these majors are a start in determining the validity of the results.

Due to the nature of the research, this being the first effective connectivity of the whole brain from rsfMRI, new results that don't match existing literature are progress but require fair and humble evaluation to determine their validity.

To mitigate the limitations arising from methodological inexperience, each result was compared to existing literature wherever possible. Several findings showed strong agreement with prior research in fMRI analysis and whole-brain connectivity. These include: the detrimental effect of infra-slow temporal fluctuations on reconstruction accuracy; the predominance of net excitatory connectivity across long-range cortical links; and the characteristic opposition between higher-order associative and lower-order sensory networks. Successfully reproducing these established large-scale features provides initial validation of the proposed inference approach and supports the credibility of the reconstructed effective connectivity.

6.3 Summary

This thesis set out to evaluate the feasibility of reconstructing whole-brain directed brain networks from resting-state fMRI data using linear least squares (LLS) inference. Through a combination of theoretical review, simulation studies, and empirical application, it established both the methodological validity and the biological relevance of this approach.

Systematic experiments on synthetic data demonstrated that, although single-subject reconstructions are limited by samples size and slow temporal dynamics, accurate network recovery is achievable through group-level averaging. The LLS method showed robustness to characteristic effects fMRI recordings, when analysing networks with high intrinsic connectivity and long activity correlation periods.

Application to Human Connectome Project rs-fMRI dataset produced, for the first time, a whole-brain linear effective connectivity for the 333 regions that revealed directed interactions consistent with known resting-state networks. The results highlighted the existence of net excitatory long-range couplings and the opposing systems of higher and lower order systems. The subsequent delay-based modelling further improved explanatory power, revealing that derivative coupling terms contribute meaningfully to the temporal coordination of brain activity.

In summary, this work provides a comprehensive evaluation of LLS-based inference for reconstructing directed neural interactions, demonstrating its capacity to produce an interpretable, data-driven effective connectome of the human brain. The findings bridge synthetic validation and empirical application, establishing a foundation for future research into dynamic and causal brain network modelling.

6.4 Future Work

There are several directions through which the current study could be extended. The most direct improvement would be to acquire longer time-series recordings or employ imaging techniques with higher temporal resolution. This enables reliable single-subject effective connectivity (EC) estimates. Such capability would open valuable directions for investigating inter-subject variability, individual brain “fingerprinting,” and personalised network characterisation.

This thesis established a baseline linear discrete model of brain connectivity, which was shown to capture key aspects of macroscopic brain dynamics. Future work could extend this framework to continuous time models, as demonstrated by Gilson et al. (2018), or incorporate nonlinear coupling functions to better represent underlying neuronal mechanisms. For instance, Kinsey et al. (2024) compared spatial patterns derived from linear and nonlinear models, revealing unique nonlinear dynamics within the default-mode and control networks that were disrupted in schizophrenia, highlighting the diagnostic potential of richer model formulations.

Ultimately, the goal of advancing inference techniques is to support clinical translation. The ability to derive robust single-subject EC maps from safe, non-invasive fMRI recordings could enable the identification of characteristic network signatures of neurological or psychiatric disorders and inform targeted, connectivity-based interventions.

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